— Supplementary Materials —

Scaling the assessment of associative ability: Development and validation of a multiple-choice compound remote associates task

Author names blinded for review

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Instructions for reproducing the analyses

In acknowledgement of recent calls for more transparent and fully reproducible research practices across virtually all sub-disciplines of psychology (Condon, Graham, & Mroczek, 2017; Gilmore, Cole, Verma, Van Aken, & Worthman, 2020; Grahe, 2018; Hagger, 2019; Klein et al., 2018; Mellor, 2021; Tackett & Miller, 2019), the supplemental materials for this project are extensive. To reproduce this file, download the zip folder under 'OSF Storage' from the OSF project page (tinyurl.com/2a8vx6n8). Open it on your computer, making note of the location where it is stored (and relocating, if you choose). Once the data have been downloaded, it should be possible to reproduce the published results by using the `R` code in the ".Rnw" version of this document (this includes several analyses — code chunks — which have been suppressed in the compiled PDF version for the sake of readability).

The code can either be executed manually in `R` or by compiling the file with Knitr (Xie, 2015) or Sweave (Leisch, 2002) (some editing of the text passages may be required once the file is opened in RStudio or text editing software). The current PDF version was compiled using Knitr in RStudio (Gandrud, 2015) on January 1, 2023. Those who do not want to use RStudio can open the .Rnw file with a text editor and run the code chunks that begin with "<<" and end with "@". The code mainly uses functions in base `R` 4.2.2 (R Core Team, 2021), psych version 2.2.5 (Revelle, 2021), and a few other packages; the names and version numbers are listed in Table S1.

Package	Version
corrplot (Wei & Simko, 2021)	0.92
dplyr (Wickham, François, Henry, & Müller, 2021)	1.0.9
ggeffects (Lüdecke, 2018)	1.1.3
gt (Iannone, Cheng, & Schloerke, 2021)	0.6.0
gtsummary (Sjoberg, Whiting, Curry, Lavery, & Larmarange, 2021)	1.6.1
kableExtra (Zhu, 2021)	1.3.4
knitr (Xie, 2015)	1.39
lme4 (Bates, Mächler, Bolker, & Walker, 2015)	1.1-30
mirt (Chalmers, 2012)	1.36.1
psych (Revelle, 2021)	2.2.5
rio (Chan, Chan, Leeper, & Becker, 2021)	0.5.29
stringr (Wickham, 2019)	1.4.0
tidyverse (Wickham et al., 2019)	1.3.1
xtable (Dahl, 2016)	1.8-4

Table S1: `R` packages used and version numbers

To get started reproducing the analyses in this file, open the file named 'SupplementalMaterials.Rnw' (the title will include the last date of modification), and set your working directory to the location of the files. There are many ways to do this — an easy one is to adjust line 97 (wdpath = ...) in the .Rnw file to match the location of the directory on your own computer.

1 Study 1

The following code loads the raw data from the [**blinded for review**] for this study. Most of the data for this study are available for download and re-use (see the compressed folder labeled 'Study1data' on OSF (tinyurl.com/2a8vx6n8) — these include all data related to CRA/associative ability items. However, as the validation data are part of a much larger data collection project, some of these data have been withheld temporarily with the expectation that they will be made available subsequently (consistent with prior data sharing related policies for this project). The code below shows how the complete data were loaded, described, and reduced down to a more manageable size for saving and subsequent analyses/sharing. See [**blinded for review**] for a general overview of the data collection methods. Note that the data used in Study 1 was collected between 07 February 2017 and 25 June 2022.

```
# First load the dataset used for the analyses.
load(paste(wdpath, "data/study1/data07feb2017thru12dec2022.rdata", sep = ""))
fullSample <- data07feb2017thru12dec2022
rm(data07feb2017thru12dec2022)
gcQuiet()

dim(fullSample)
count_answered <- describe(fullSample$count_answered)

pdf(paste(wdpath, "figure/counts.pdf", sep = ""), onefile = TRUE)
hist(fullSample$count_answered, 250, xlab="Items Answered",
     xlim = c(0,250), ylim = c(0,200000), main = "")
dev.off()

# subset the data
fluencyAll <- table(fullSample$english)
CRAdata <- fullSample %>% dplyr::filter(english == "Well" | english == "VeryWellFluentNative")
dim(CRAdata)
rm(fullSample)
gcQuiet()

# based on age
CRAdata <- subset(CRAdata, CRAdata$age > 13)
dim(CRAdata)

# create table objects for printing further down
countryCount <- table(CRAdata$country)
ncountry = sum(countryCount != 0)
USperc = as.numeric(countryCount["USA"])/nrow(CRAdata)

ethnic <- data.frame(table(CRAdata$ethnic))
ethnic$Percentage <- round(ethnic$Freq/sum(ethnic$Freq), 3)*100
colnames(ethnic) <- c("Ethnicity/Race", "Participants", "Percentage")

education <- data.frame(table(CRAdata$education))
education$Percentage <- round(education$Freq/sum(education$Freq), 3)*100
colnames(education) <- c("Educational Level", "Participants", "Percentage")

jobstatus <- data.frame(table(CRAdata$jobstatus))
jobstatus$Percentage <- round(jobstatus$Freq/sum(jobstatus$Freq), 3)*100
colnames(jobstatus) <- c("Job Status", "Participants", "Percentage")

IRTscores <- c("SPI_135_27_5_Impulsivity_score", "SPI_135_27_5_Adaptability_score",
               "SPI_135_27_5_SelfControl_score", "SPI_135_27_5_Compassion_score",
               "SPI_135_27_5_Creativity_score", "SPI_135_27_5_Charisma_score",
               "SPI_135_27_5_Perfectionism_score", "SPI_135_27_5_Conformity_score",
               "SPI_135_27_5_Order_score", "SPI_135_27_5_EmotionalStability_score",
               "SPI_135_27_5_Irritability_score", "SPI_135_27_5_EmotionalExpressiveness_score",
               "SPI_135_27_5_Intellect_score", "SPI_135_27_5_Humor_score",
               "SPI_135_27_5_Sociability_score", "SPI_135_27_5_Conservatism_score",
               "SPI_135_27_5_ArtAppreciation_score", "SPI_135_27_5_Authoritarianism_score",
               "SPI_135_27_5_Trust_score", "SPI_135_27_5_Honesty_score",
               "SPI_135_27_5_AttentionSeeking_score", "SPI_135_27_5_WellBeing_score",
               "SPI_135_27_5_SensationSeeking_score", "SPI_135_27_5_Introspection_score",
               "SPI_135_27_5_Anxiety_score", "SPI_135_27_5_Industry_score",
               "SPI_135_27_5_EasyGoingness_score", "ICAR_LN_IRT_score",
               "ICAR_MR_IRT_score", "ICAR_R3D_IRT_score",
               "ICAR_VR_IRT_score", "ICAR_FA_IRT_score",
               "ICAR_PR_IRT_score")
demoShort <- c("RID", "servertime", "age", "sex", "major", "occupation")
Study1items <- c(demoShort, itemsICAR, itemsSPI, IRTscores)
```

```
CRAdata <- subset(CRAdata, select = c(Study1items))
dim(CRAdata)

# load the item information with the text and labels
load(paste(wdpath, "data/CRAitemContent.rdata", sep = ""))
# and a file used to describe the sample by country
load(paste(wdpath, "data/study1/countrykey.rdata", sep = ""))

CRAitemData <- subset(CRAdata, select = c(servertime, q_14003:q_14466))
save(CRAitemData, CRAitemContent, file = paste(wdpath, "data/study1/CRAdataStudy1version1.rdata", sep = ""))

save(count_answered, countryCount, countrykey, CRAdata, CRAitemContent, ethnic, education, jobstatus,
     fluencyAll, itemsSPI, jobstatus, keys.list, ncountry, USperc,
     file = paste(wdpath, "data/study1/CRAdataStudy1version2.rdata", sep = ""))
```

1.1 Descriptive statistics for participants in the sample

The code below generates descriptive statistics for the sample. Some of the code in this sub-section (for reproducing the descriptive statistics) and in sub-section 1.7 (for identifying the correlations with other traits) requires access to the complete data file (version 2) ; the remaining code is reproducible with version 1 only.

```
load(paste(wdpath, "data/study1/CRAdataStudy1version2.rdata", sep = ""))

CRAdata <- CRAdata[CRAdata$servertime < as.POSIXct("2022-06-26 12:00:00", tz="UTC"),]
dim(CRAdata)

## [1] 1323480    1288

occupation = table(CRAdata$occupation)
occupations <- data.frame(sort(occupation, decreasing = TRUE))
colnames(occupations) <- c("Jobs", "Participants")
nocc = sum(occupation != 0)
pocc = sum(table(CRAdata$occupation))

ages <- describe(CRAdata$age)

sex.t = CRAdata %>%
  group_by(sex) %>%
  dplyr::summarize(count = n()) %>%
  mutate(percent = 100*count/sum(count))
```

The total sample included 1,845,491 participants who completed all or part of the full inventory in exchange for information about their personalities. Figure S1 provides a distribution of the number of responses provided by each participant (m = 128.4; sd = 81.9; Mdn = 123).

From the full sample, participants were excluded if they self-reported that they did not speak English well (n = 97,006) or at all (i.e., needed translation; n = 19,474), or if they did not indicate their English language fluency (n = 177,728). This left a sample of **1,323,480 participants** who spoke English well (n = 723,166) or very well/fluent (n = 828,117). This sample had an average age of 26.5 (sd = 12.6; Mdn = 22; minimum = 14; maximum = 90). Approximately 69% self-identified as female, 30% as male, and 0.35% as other (1.28% did not respond to the item about gender).

With respect to geographic distribution, 232 nation states were represented and approximately 23.63% of respondents were from the U.S. This included representation from all regions of the U.S. at a level similar to the distribution of residents, as indicated by moderate to high correlations between the distribution of the sample and the distribution of U.S. population (.47 using smaller 5 digit ZIP code areas and .81 using larger 3 digit ZIP code areas). Figures S2 and S3 show the geographic distribution of the sample within the U.S. and by country, respectively. Table S2 shows the number of respondents per country for those with more than 1,500 respondents.

Tables S3 and S4 show the distribution of the sample by ethnicity/race and educational attainment level, respectively. Table S5 shows the distribution of the sample by employment status. Participants who reported being employed or retired were prompted to indicate their occupation; 394,044 participants responded to this prompt, representing a total of 1,027 jobs.

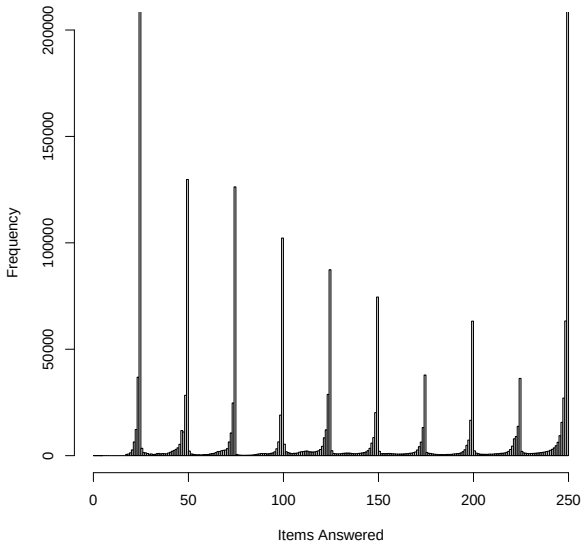


Figure S1: Distribution of responses per participant

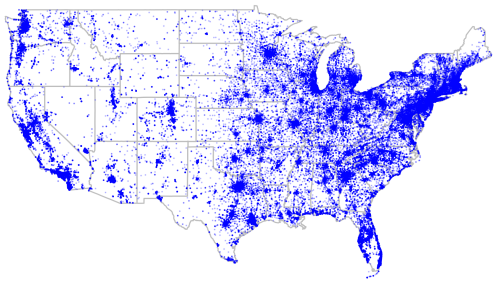


Figure S2: Full sample by U.S. Zip Code Tabulation Area

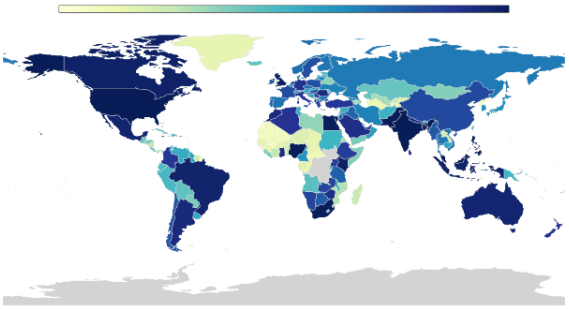


Figure S3: Full sample by country

	Participants	Population Rank	English Speaking Rank
USA	355098	3	1
India	281116	2	3
Philippines	127289	13	4
Pakistan	76594	6	13
Nigeria	67332	7	2
South Africa	33439	25	19
Malaysia	31303	45	21
Bangladesh	31293	8	12
United Kingdom	30396	21	5
Kenya	29835	30	32
Egypt	27345	14	9
Canada	25164	38	8
Indonesia	17127	4	NA
Brazil	14591	5	28
Australia	13970	53	11
Lebanon	12258	110	70
Jamaica	10953	139	59
Mexico	10899	10	20
Trinidad and Tobago	9997	152	75
Morocco	9107	39	43
Ghana	8939	49	15
Zimbabwe	8882	70	26
Romania	8644	59	38
Saudi Arabia	8040	41	NA
Argentina	7647	32	58
Nepal	7003	48	78
Germany	6261	16	6
New Zealand	5388	124	47
Algeria	5324	35	61
Sri Lanka	5192	58	24
United Arab Emirates	4909	93	NA
Italy	4906	23	18
Ethiopia	4732	12	NA
Uganda	4665	34	7
Turkey	4557	19	25
Greece	4359	81	39
Serbia	4348	96	NA
Namibia	4268	142	90
Zambia	4261	66	67
Colombia	3972	29	64
Singapore	3738	114	46
France	3608	22	10
Albania	3525	138	NA
Sweden	3475	88	31
Netherlands	3360	67	22
Poland	3293	37	23
Ireland	3103	122	45
China	2934	1	30
Hungary	2827	90	65
Botswana	2760	143	82
Croatia	2709	128	60
Chile	2606	65	71
Jordan	2597	91	56
Tanzania	2457	24	49
Mauritius	2448	157	96
Portugal	2357	87	54
Iraq	2157	36	27
Norway	2148	117	44
Burma - Union of Myanmar	2102	26	62
Kuwait	2039	129	NA
Finland	1974	115	50
Czech Rep	1940	86	57
Spain	1934	31	29
Puerto Rico	1926	133	66
Bulgaria	1855	104	68
Russian Federation	1693	9	16
Vietnam	1677	15	NA
Brunei	1656	175	101
Macedonia	1553	146	NA
Belgium	1538	79	35
Iran	1533	18	NA
Lithuania	1511	140	76

Note: Ranks based on 2022 estimates from Wikipedia.

Table S2: Countries with 1500 or more respondents

Ethnicity	Participants	Percentage
White	140,991	73%
African American	10,443	5.4%
Two Or More Ethnicities	10,690	5.5%
Mexican	9,580	5%
Other Hispanic	5,392	2.8%
Chinese	2,285	1.2%
Other Asian	1,959	1%
Puerto Rican	1,812	0.9%
Indian	1,817	0.9%
Filipino	1,710	0.9%
Native American	1,532	0.8%
Korean	1,067	0.6%
Cuban	576	0.3%
Japanese	511	0.3%
Other Pacific Islander	243	0.1%
Native Hawaiian	191	0.1%
Alaskan Native	131	0.1%
Other	1,519	0.8%

Table S3: Race/Ethnicity among U.S. participants in the Study 1 sample

Educational Level	Participants	Percentage
Less than 12 years	137,467	10.5%
High school graduate	202,392	15.5%
Currently in college/university	357,217	27.4%
Some college/university, but did not graduate	72,536	5.6%
Associate Degree	35,646	2.7%
College/university degree	249,746	19.2%
Currently in graduate or professional school	54,222	4.2%
Graduate or professional school degree	194,874	14.9%

Table S4: Educational attainment level in Study 1 sample

Employment Status	Participants	Percentage
Currently a student	527,522	43.9%
Employed	450,353	37.4%
Not employed	107,566	8.9%
Not employed, seeking work	64,183	5.3%
Homemaker	21,739	1.8%
Retired	31,522	2.6%

Table S5: Employment status in the Study 1 sample

1.2 Descriptive statistics for the 67 new CRA items

Next, we begin examining the properties of the CRA items. It is important to note that these items were administered infrequently due to the planned missingness design of the data collection, so we need to better specify the descriptive statistics of the CRA items by themselves. First, we subset the data, describe it, and then look at the number of pairwise administrations. This latter characteristic is important for subsequent evaluation of the structure of the set. We also get the tetrachoric correlation matrix at this stage; note that smoothing was necessary as the matrix was not positive definite.

```
load(paste(wdpath, "data/study1/CRAdataStudy1version1.rdata", sep = ""))

CRAitemData <- CRAitemData[CRAitemData$servertime < as.POSIXct("2022-06-26 12:00:00", tz="UTC"),]
dim(CRAitemData)

## [1] 1323480     181

CRAitemSet67 <- subset(CRAitemData, select = c(q_14003:q_14197))
dim(CRAitemSet67) # only the 67 item subset being validated

## [1] 1323480      67

descCRAitems <- psych::describe(CRAitemSet67)
rm(CRAdata)
gcQuiet()

administrations <- describe(descCRAitems$n)

pwiseAdminsCRAitems <- pairwiseCount(CRAitemSet67, diagonal = TRUE)
pwiseAdminsCRAitemsVec <- as.vector(pwiseAdminsCRAitems[lower.tri(pwiseAdminsCRAitems)])
pwiseAdminsCRAitemsDescribe <- psych::describe(pwiseAdminsCRAitemsVec)
n_obs <- pwiseAdminsCRAitemsDescribe$mean

tetchor_matrix67 <- tetrachoric(CRAitemSet67, smooth = TRUE)
```

```
cor_matrix67 <- tetchor_matrix67$rho
```

The 67 items in this set have been administered an average of 15,778 times each (sd = 2,670.1; Mdn = 14,867), with a minimum of 14,290. However, these values are vastly larger than the pairwise administrations among all 2,211 possible pairs, as these had a mean of only 528.4 (sd = 153.5; Mdn = 511). This circumstance — much higher administration counts per item than pairwise administrations — is due to the relatively small number of items administered to each participant (often only two). The data collection protocol on the **[blinded for review]** only dedicates about 10% of survey content to cognitive abilities assessment, and much of this is taken up by the administration of previously validated scales, i.e., the four original item types described in Condon and Revelle (2014). Importantly, the evaluations of validity reported in later sections of this work (e.g., discriminant and convergent validity) benefit from the statistical power of the larger number of individual administrations against other criteria. Evaluations of the correlational structure among items do not, and this prompts consideration of whether the current number of observations is adequate for the structural evaluations conducted here.

Various "rules of thumb" have been proposed regarding statistical power for exploratory factor analysis; see Kyriazos (2018) for an excellent summary. In addition to the typical considerations affecting power (e.g., thresholds for alpha and beta, anticipated effect sizes, and number of observations), issues affecting power in EFA include the number of measured variables, the number of dimensions being extracted (i.e., the factors or components), and the observed orientation of the variables to those dimensions (i.e., the communalities and the loadings). This complexity contributes to the appeal of using generic rules to estimate power for EFA, but it should be noted that these rules are often quite arbitrary. The categories provided by Comrey and colleagues (Comrey, 1973), for example, are overly simplistic — they suggest grading sample sizes independent of contextual factors: very poor (N=50), poor (N=100), fair (N=200), good (N=300), very good (N=500), and excellent (N=1,000).

More nuanced rules are often based on detailed simulations (Dimitrov, 2014; Guadagnoli & Velicer, 1988). The most applicable for EFA specifically are framed around the ratio of observations (N) to variables (p; Dimitrov, 2014; Nunnally, 1994). Recommendations range from N/p = 1 for larger samples to 5 (Gorsuch,

1983) or even 10 (Nunnally, 1994; Wang & Wang, 2019) when samples sizes are small (i.e., Ns of roughly 100 might only be considered marginally adequate for small numbers of variables). A particularly useful recommendation from Tinsley and Tinsley (1987) proposes 5 to 10 observations per variable up to N=300, beyond which this proportion can be relaxed (DeVellis & Thorpe, 2021; Kyriazos, 2018). The data used here comfortably exceeds these recommendations — again, the mean and median are 528.4 and 511, respectively, and the minimum is 327.

1.3 Dimensionality of the 67 new CRA items

The next step is to evaluate the unidimensionality of the items, as this is a key assumption to fulfill when using item response theory-based procedures (and we want to use these in subsequent analyses, if possible). We begin by calling `vss()` to generate several test statistics for various factor solutions.

```
fitstats <- psych::vss(cor_matrix67, n = 4, n.obs = n_obs, rotate = "oblimin", diagonal = FALSE, fm = "mle", plot = FALSE)

fit_table <- fitstats$vss.stats

fit_table <- fit_table %>%
  mutate(Factors = seq(1:4)) %>% # Look at fit indices for 1-4 factor solutions
  relocate(Factors)

fit_table <- fit_table %>%
  select(Factors, chisq, fit, RMSEA, BIC, SRMR) # Selecting preregistered fit indices

fit_table <- fit_table %>%
  mutate(Map = fitstats$map) %>%
  dplyr::rename(VSS = fit,
         Chisq = chisq)

# suppressMessages(suppressWarnings(fa.parallel(cor_matrix67)))
# Parallel analysis suggests that the number of factors =  2
# and the number of components =  1
```

Factors	Chi Square	VSS	RMSEA	BIC	SRMR	MAP
1	195483	0.90	0.41	182040	0.082	0.016
2	194186	0.81	0.42	181157	0.074	0.015
3	193125	0.79	0.42	180503	0.069	0.015
4	192199	0.78	0.43	179979	0.065	0.015

Note: VSS shown here reflects complexity of 1. SRMR is the standardized root mean square residual. MAP is Velicer's Minimum Average Partial criterion.

Table S6: Fit indices for the 1 to 4 factor solutions with 67 items

We evaluate the fit for solutions with up to 4 factors, and then create Table S6 with the output. Table S6 displays model fit indices for solutions with one through four factors. The optimal number of factors is inconsistent across fit indices. Support for the single-factor model is indicated by the VSS (Very Simple Structure) criterion, which suggests that extracting one factor yields the most interpretable results. The SRMR suggests at least adequate fit (i.e., less than .08) for all solutions. The chi-square value, BIC (Bayesian Information Criterion), and MAP (Velicer's Minimal Average Partial) criterion were minimized in models with a higher number of factors. RMSEA values were high for all models, far exceeding the standard threshold for adequate fit (i.e., of less than .08). Reasons for notably poor fit across all solutions as indicated by the RMSEA are unclear, but recent evidence suggests that the RMSEA — in addition to other popular fit indices — is ill-suited for use in exploratory factor analysis because it almost always leads to over-factoring (Montoya & Edwards, 2021). Consequently, RMSEA values are reported for the purpose of adhering to the preregistered plan of analysis but are not used to guide interpretation regarding the factor structure of the CRA items. Figure S4 shows the results of parallel analysis suggesting two factors or one principal component.

The `omega()` function in the psych package is used to show coefficient ω_h and (for internal consistency) α and ω_t.

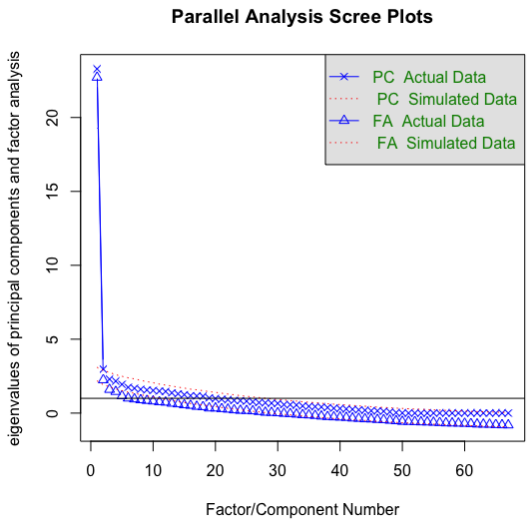


Figure S4: Parallel analysis based on the 67 CRA items using principal components (PC) and factor analysis (FA)

```
omega_results <- psych::omega(cor_matrix67, plot = FALSE)
omega.h <- round(omega_results$omega_h,2)
omega.alpha <- round(omega_results$alpha,2)
omega.t <- round(omega_results$omega.tot,2)
```

The items have high internal consistency in this data set ($\alpha = 0.97$; $\omega_t = 0.97$). Similarly, the omega general value ($\omega_h = 0.87$) suggests strong unidimensionality in the CRA items. To help visualize this, the next code chunk plots the correlations among items after they have been sorted by factor loading, with a one factor model specified (sorting by factor loading is only done to organize the presentation of items). We also extract the empirically-observed confidence intervals around the tetrachoric correlations here.

The correlations are shown in Figure S5. The plot provides clear evidence of a strong general factor (average $r = 0.326$), though there are a small number of correlations between items with point estimates below 0 — as low as -0.093 — indicated with pale red tones in the lower right corner of the figure. The range of the 95% confidence intervals around these correlations (i.e., the range from the high to low end of the confidence interval) had a mean of 0.15 (minimum = 0.073 and maximum = 0.171). Of the 2,211 pairwise correlations, 106 had confidence intervals overlapping with zero; 2 of the remaining 2,105 had point estimates below 0. The correlations among these "significantly" negative associations are -0.09 (q_14058 and q_14082) and -0.09 (q_14096 and q_14148). The content of these items can be seen in Table S7. It is unclear why these are negatively associated. One possibility is that there is a cohort effect that makes knowledge/familiarity of the words in each pair relatively incompatible. A second possibility is that one (or both) of the items in each pair prompts a near alternative among the compound words in the matching item. Further research into this topic is warranted, but none of these items are dropped as they demonstrate adequate psychometric properties in other respects.

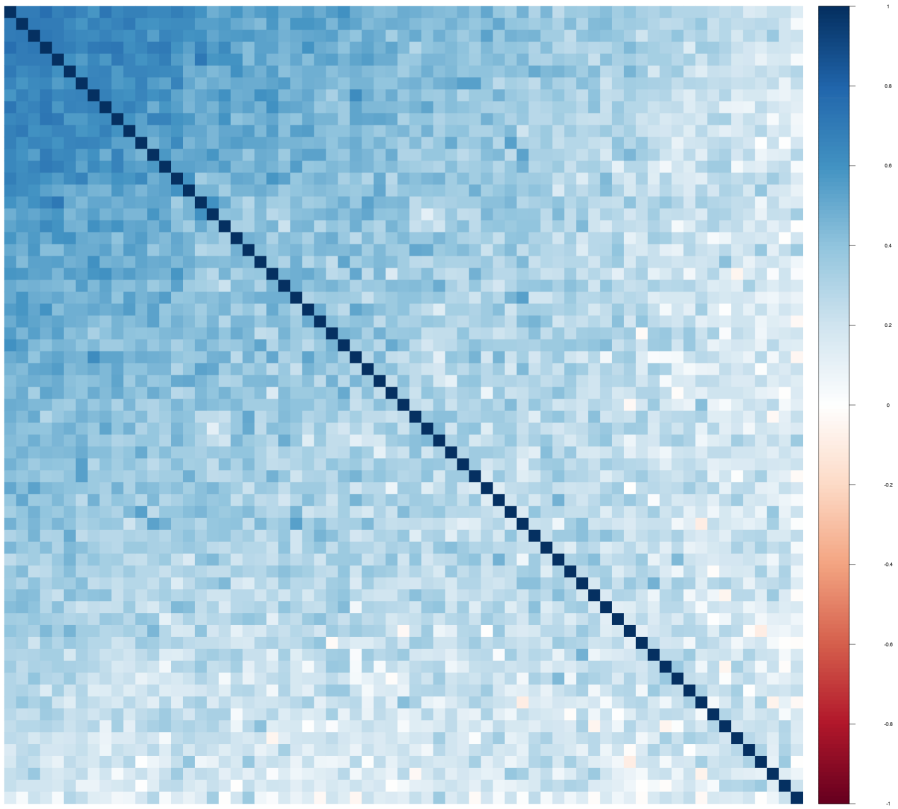


Figure S5: Correlations among all 67 CRA items sorted by general factor loadings.

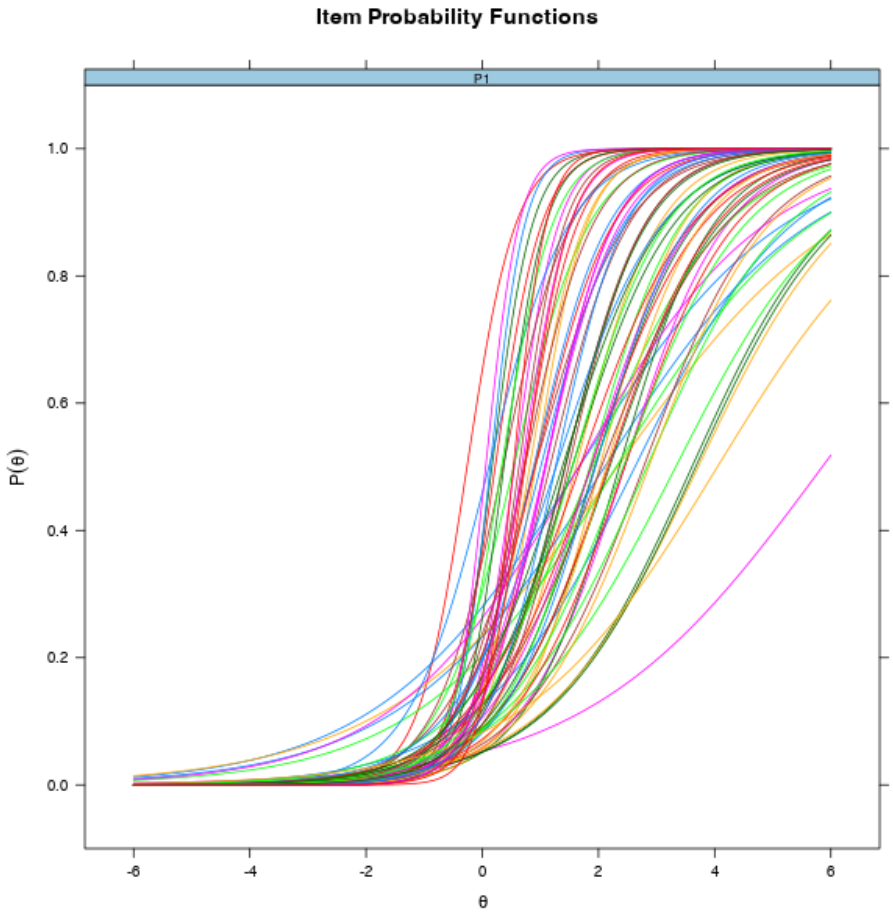


Figure S6: Item probability functions for all 67 CRA items together.

1.4 IRT-based analyses of the 67 new CRA items

Supported by the evidence of unidimensionality, the item pool was evaluated using item response theory procedures. In the chunk below, we extract the IRT parameters, plot the information curves for the items and the set, and evaluate the range of item difficulties.

Figure S6 shows the probability functions — that is, the probability of correct/incorrect responses across the range of difficulties — for all the items together in one plot. Figure S7 shows the probability functions for all 67 items separately and labeled. Together, these figures are very useful for evaluating the properties of the CRA items individually and as a set.

Figure S6 shows that most items have difficulties greater than zero ($m = 1.61$; $Mdn = 1.55$). As mentioned above, difficulty values (in this model) reflect the location of the intercept where the probabilities of correct and incorrect responses are equal. Only 1 of the 67 items have difficulty coefficients below zero, and this indicates that most items are generally quite hard. Note that the mean proportion of correct responses for each item is highly (negatively) correlated with item difficulty (r = -0.84), as expected. Given the content of these items, moderately high difficulties are desirable, but items with excessively high difficulty would not be particularly useful as there will be little variance across respondents (nearly all respondents fail to identify the correct answer). There are two items with difficulties more than 4 standard deviations above the mean ability level: item q_14043 (difficulty = 5.85) and item q_14175 (difficulty = 4.05).

Figures S6 and S7 also demonstrate that most of the items are adequately discriminant ($m = 1.48$; Mdn

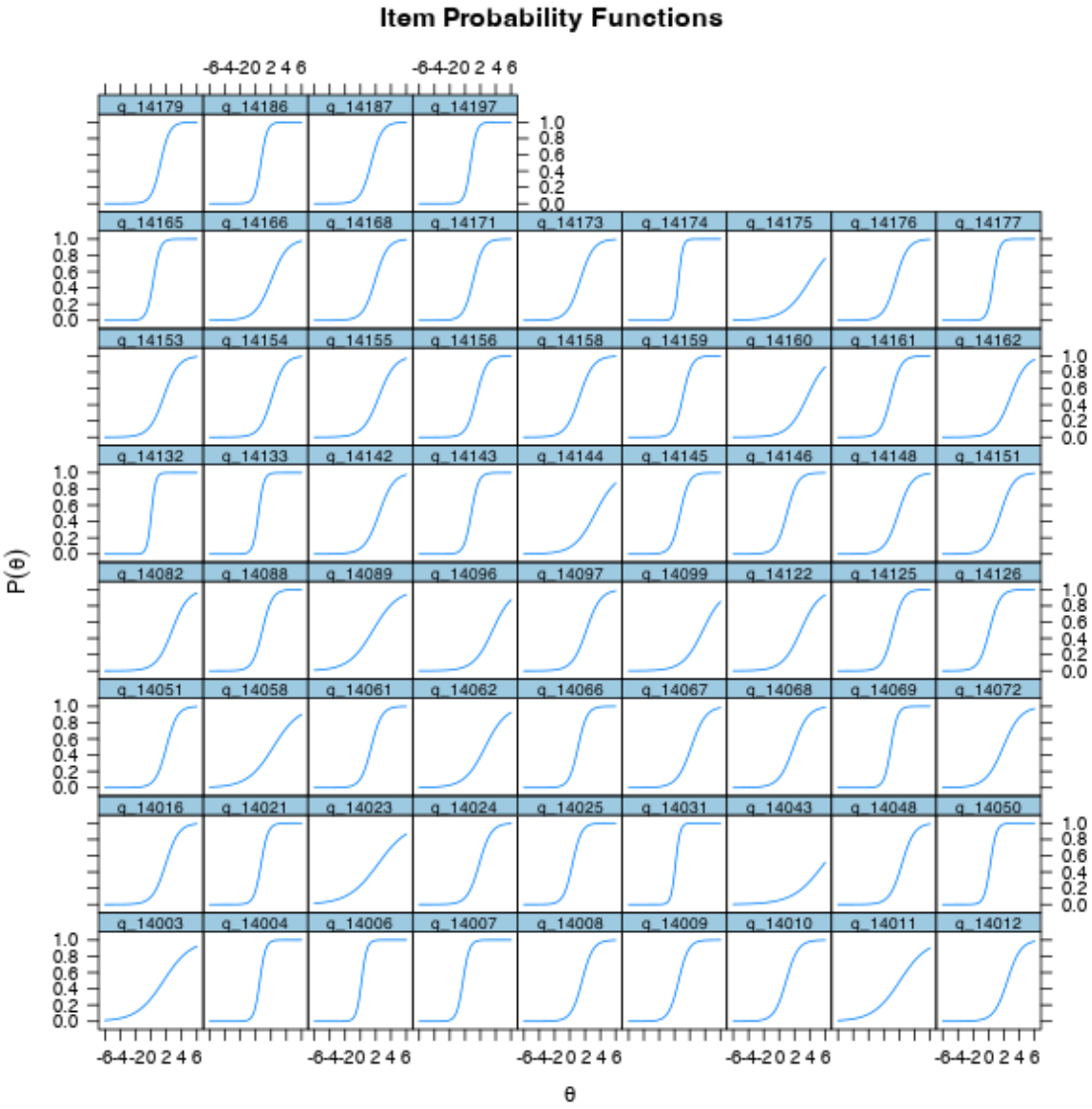


Figure S7: Item probability functions for all 67 CRA items.

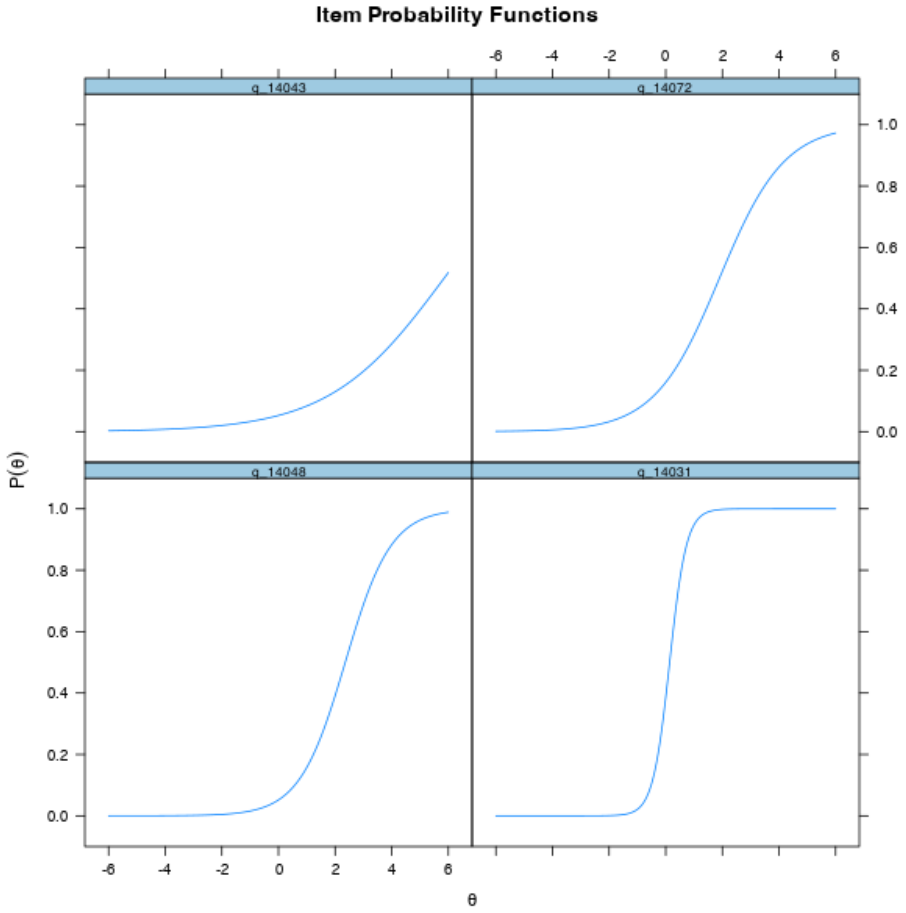


Figure S8: Examples of varying discrimination values among the CRA items.

= 1.28), though there are a few exceptions. This implies that the items are able to discriminate well between individuals with relatively similar levels of associative ability around the difficulty intercept, as a small increase in associative ability in this range substantially increases the probability of correct responding. This is depicted visually by the slope of the item probability functions. Of the 67 items, two have discrimination parameters near .5, a value that is commonly deemed a lower bound of acceptability (An & Yung, 2014; Debeer & Janssen, 2013). Note that it may also be reasonable to drop a few other items with relatively low discriminations, but there is a case to be made for keeping items that cover a wide range of discriminations, especially if they are all above an acceptable threshold (Cheng & Liu, 2015). This argument mainly rests on the increased flexibility provided by a diverse item set for computer adaptive testing applications.

For reference, Figure S8 highlights the discrimination for several items as examples. Item q_14043 has among the lowest discriminations of all 67 items at 0.49. In the top right, item q_14072 has a low but adequate discrimination of 0.87. Items q_14154 and q_14031 have the median and maximum discrimination values, respectively (1.23 and 3.37).

1.5 Identifying and reporting on the final set of (65) new CRA items

As suggested in the prior section, there are two items close to the lower threshold of acceptable discrimination values (slightly above and/or at 0.5). These items are q_14043 and q_14023. Several other items have discriminations between 0.55 and 0.65 — these are items q_14011, q_14003, q_14175, and q_14058. Note that

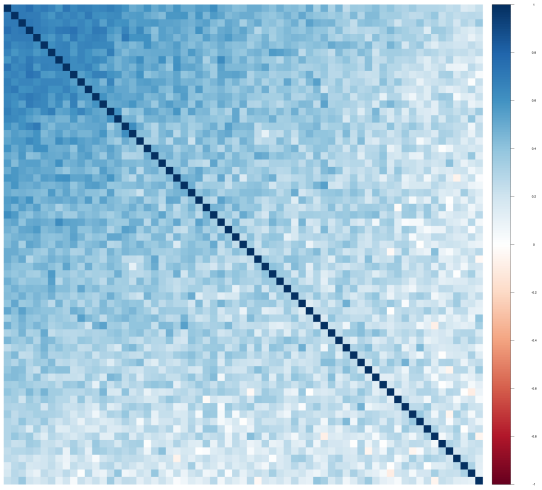


Figure S9: Correlations among the final CRA items sorted by general factor loadings.

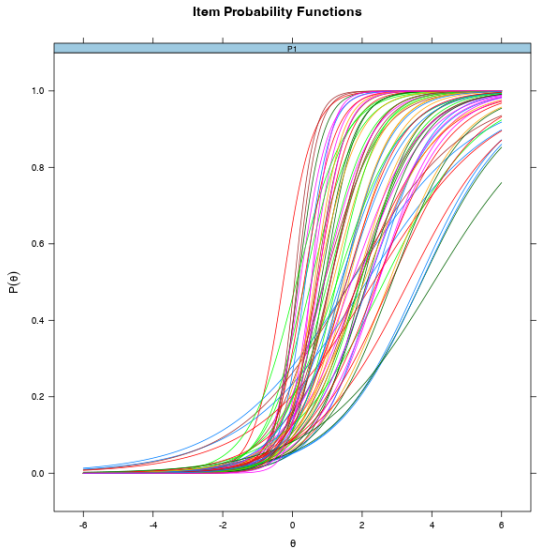


Figure S10: Item probability functions for the final CRA items together.

3 of the latter set of items have a correct answer beginning with a common letter ("s"), and it may be that this reflects a somewhat effective guessing strategy to the detriment of the discrimination of those items. All of these six items are listed below:

- q_14043 — **prompt:** front side bed; **response options:** o,p,w,a,t,l; **answer:** l (lake)
- q_14023 — **prompt:** card board keeper; **response options:** d,t,a,o,s,r; **answer:** s (score)
- q_14011 — **prompt:** line rocket box; **response options:** s,m,p,d,v,o; **answer:** s (sky)
- q_14003 — **prompt:** bed sheet wide; **response options:** m,y,s,v,k,a; **answer:** s (spread)
- q_14175 — **prompt:** line water war; **response options:** p,f,s,m,n,k; **answer:** f (front)
- q_14058 — **prompt:** bed jet blood; **response options:** f,m,b,s,r,w; **answer:** s (stream)

Ultimately, it was decided to drop the least discriminating of these items (q_14043 and q_14023) and retain the rest. The decision to retain the latter 4 (q_14011, q_14003, q_14175, and q_14058) was based on the case for retaining as many items as possible for the sake of potentially developing a CAT at some future point. However, the decision to retain these items should be reevaluate in subsequent research, especially if contradicting evidence is found in other samples (that is, if the items continue to have borderline discrimination values).

In the remainder of this section, we re-run the code from the prior subsections after removing the 2 undesirable items; this leaves only 65 items in total. The new analyses generate updated versions of the correlation plot and item probability functions; see Figures S9, S10, and S11.

In Table S7, we present the results numerically, overlapping with the figures to some extent. This includes coefficients for the item difficulties (the location of the 50/50 probability of a correct response) and discrimination parameters (the slopes), as well as the mean proportion of correct responses and the standard error of the means.

Figure S12 shows the joint information for the final set of CRA items across the range of difficulty along with the standard error. As the standard error is akin to the inverse of reliability (more specifically, reliability is 1 minus the square of the standard error over the standard deviation), Figure S12 indicates that the test

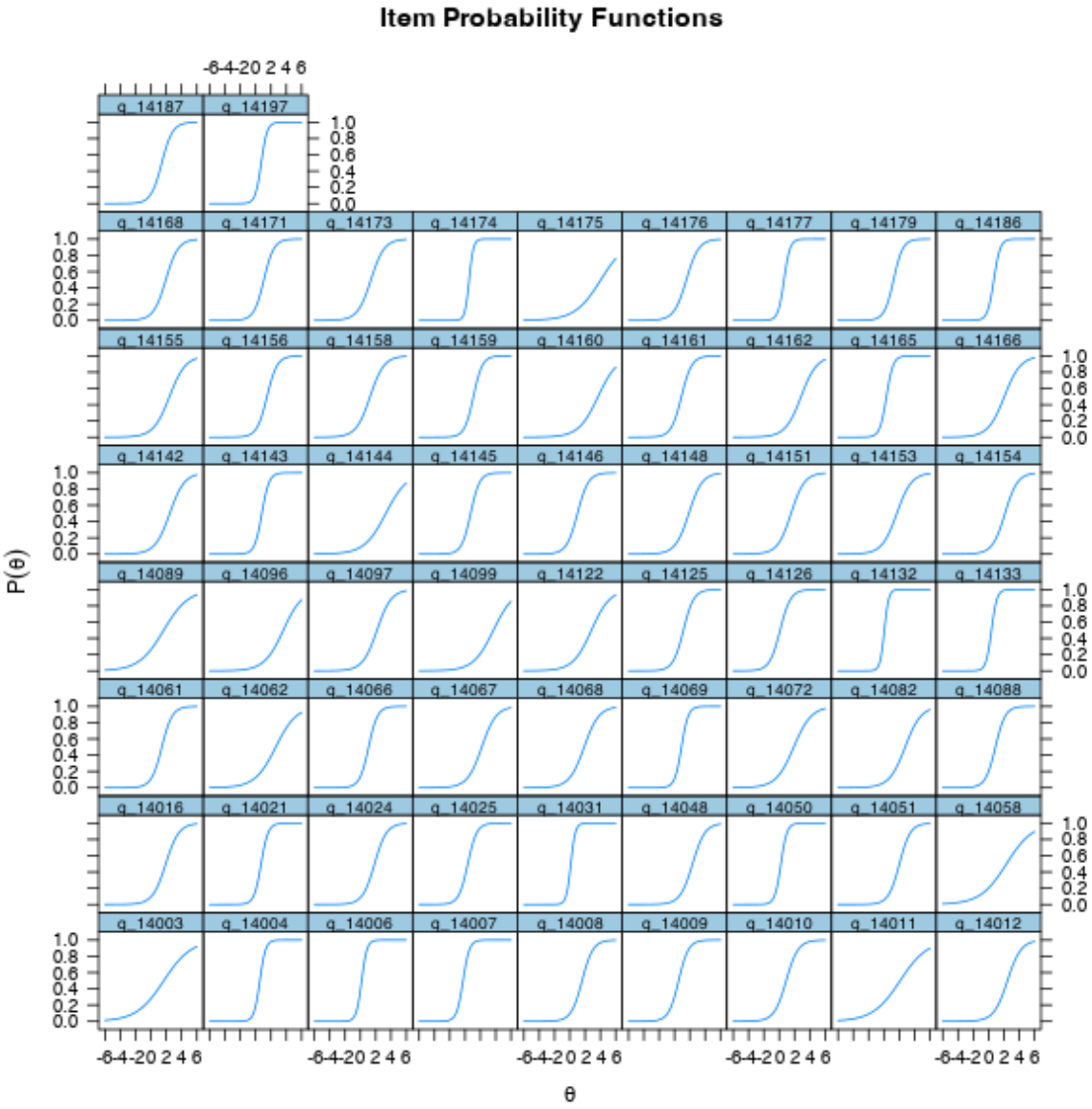


Figure S11: Item probability functions for the final CRA items.

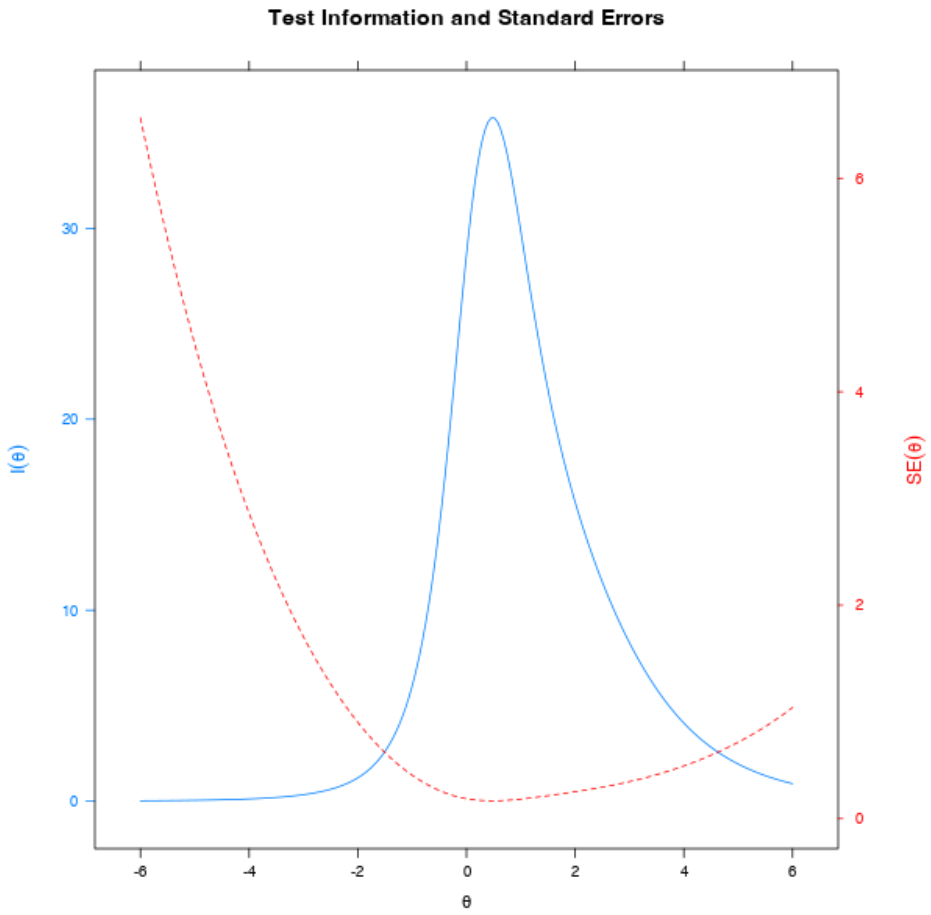


Figure S12: Test information (blue) and standard error (red) functions for the new CRA items.

is particularly reliable from the range of difficulty of about -0.5 sd to +3 sd. The marginal reliability across all respondents in these data is high at 0.91 (Thissen & Wainer, 2001).

The final step of the IRT analyses for the new items is to generate latent trait scores for respondents based on the item parameters from this 2PL model. The next code chunk does just that, appending the scale scores to the end of our data frame based on responses to the final set of new items.

```
# use the model to generate IRT scores
CRAitemSetFinal <- CRAitemSetFinal[,1:65]

empty_rows = which(is.nan(rowMeans(CRAitemSetFinal, na.rm = TRUE)))
CRAitemSetTemp <- CRAitemSetFinal[-empty_rows,]
scored <- fscores(object = IRT_model, response.pattern = CRAitemSetTemp)
CRAitemSetFinal[-empty_rows,c("CRA_new_IRT_score", "CRA_new_IRT_score_SE")] <- scored
save(CRAitemSetFinal, file = paste(wdpath, "data/study1/CRAitemSetFinal.rdata", sep = ""))
CRAnewItemNames <- colnames(CRAitemSetFinal)
IRT_model_ICAR_CRAnew <- IRT_model
save(IRT_model_ICAR_CRAnew, CRAnewItemNames, file = paste(wdpath, "data/study1/IRTmodelCRAnewItems.rdata", sep = ""))
```

	Stimuli 1	Stimuli 2	Stimuli 3	Solution	discrimination	difficulty	mean	se	n
q_14003	bed	sheet	wide	spread	0.56	1.68	0.29	0.004	14320
q_14004	war	thorough	fan	fare	2.64	0.55	0.32	0.004	15284
q_14006	soil	mountain	table	top	2.86	0.17	0.44	0.004	14761
q_14007	junk	grave	stick	yard	2.38	-0.26	0.60	0.003	23833
q_14008	stone	word	hole	key	1.40	1.61	0.15	0.003	15118
q_14009	cake	board	tea	cup	1.55	0.54	0.36	0.004	14586
q_14010	fire	card	life	wild	1.28	0.91	0.29	0.004	15487
q_14011	line	rocket	box	sky	0.56	2.12	0.24	0.004	14930
q_14012	line	lock	may	hem	1.13	2.46	0.09	0.002	15278
q_14016	star	pan	saw	dust	1.16	1.94	0.14	0.003	15219
q_14021	key	table	pike	turn	2.17	0.72	0.29	0.004	14659
q_14024	house	time	peace	keeper	1.28	1.85	0.14	0.002	23583
q_14025	bitter	heart	bread	sweet	1.71	0.36	0.40	0.004	14810
q_14031	sweet	throb	worm	heart	3.37	0.14	0.46	0.004	14714
q_14048	tow	foot	war	path	1.23	2.36	0.09	0.002	15061
q_14050	wart	hedge	wash	hog	2.51	0.22	0.43	0.004	15388
q_14051	point	pocket	pen	knife	1.40	2.03	0.10	0.002	14768
q_14058	bed	jet	blood	stream	0.59	2.33	0.22	0.003	14588
q_14061	sun	cloud	star	burst	1.47	1.43	0.17	0.003	14771
q_14062	field	piece	board	center	0.74	2.62	0.15	0.003	14724
q_14066	breaker	pipe	burn	wind	1.70	1.08	0.22	0.003	15200
q_14067	root	room	wire	tap	1.06	2.19	0.12	0.003	14628
q_14068	fire	air	stroke	brush	1.03	1.88	0.17	0.002	23372
q_14069	neck	bump	berry	goose	2.26	0.91	0.23	0.004	14519
q_14072	nose	neck	blue	bottle	0.87	1.90	0.19	0.003	14548
q_14082	key	bowl	line	punch	0.97	2.77	0.09	0.002	14965
q_14088	ache	pot	button	belly	1.55	0.88	0.27	0.004	14719
q_14089	roll	boat	punk	steam	0.62	1.67	0.28	0.003	23510
q_14096	snow	bird	pole	cat	0.80	3.61	0.07	0.002	14903
q_14097	wheel	load	hand	cart	1.07	2.17	0.12	0.003	15031
q_14099	play	line	book	date	0.76	3.71	0.07	0.002	15119
q_14122	child	day	place	birth	0.83	2.83	0.11	0.003	15238
q_14125	pine	sauce	crab	apple	1.47	1.16	0.22	0.003	14767
q_14126	bishop	way	rival	arch	1.45	0.11	0.47	0.004	14823
q_14132	bird	robe	tub	bath	3.51	0.06	0.48	0.004	14745
q_14133	stalk	jelly	pole	bean	2.81	0.37	0.38	0.004	15119
q_14142	child	wash	power	brain	1.04	2.46	0.10	0.002	14867
q_14143	thorough	pure	high	bred	2.12	0.82	0.26	0.004	15271
q_14144	fly	work	draw	bridge	0.72	3.33	0.10	0.002	14711
q_14145	stock	pawn	power	broker	1.70	0.73	0.31	0.003	23648
q_14146	fly	bean	cup	butter	1.41	1.01	0.26	0.004	15070
q_14148	ground	base	fire	camp	1.14	1.87	0.15	0.003	14554
q_14151	post	stock	board	card	1.14	1.55	0.19	0.003	14852
q_14153	rain	book	list	check	1.04	1.69	0.19	0.003	14898
q_14154	care	brain	grand	child	1.22	2.14	0.11	0.002	23403
q_14155	room	middle	first	class	0.95	2.42	0.12	0.003	15321
q_14156	burst	top	thunder	cloud	1.52	1.48	0.16	0.003	15476
q_14158	photo	hard	cat	copy	1.10	1.27	0.24	0.004	14757
q_14159	tail	mouth	wood	cotton	1.72	1.09	0.22	0.003	14573
q_14160	room	house	side	court	0.79	3.70	0.06	0.002	14470
q_14161	fire	nut	jack	cracker	1.49	0.91	0.27	0.004	15207
q_14162	quarter	hand	sun	deck	0.97	2.86	0.09	0.002	23070
q_14165	break	stead	ball	fast	2.29	0.33	0.40	0.004	14939
q_14166	grass	rain	breast	fed	0.94	2.05	0.16	0.003	14461
q_14168	board	plan	dance	floor	1.24	1.95	0.13	0.003	14429
q_14171	print	race	ball	foot	1.54	1.08	0.23	0.004	14537
q_14173	pitch	ball	lift	fork	1.15	1.38	0.22	0.003	14673
q_14174	hole	trot	glove	fox	3.59	0.54	0.31	0.004	14586
q_14175	line	water	war	front	0.59	4.06	0.09	0.002	14814
q_14176	fish	video	end	game	1.25	1.55	0.18	0.003	14290
q_14177	finch	smith	fish	gold	2.55	0.63	0.30	0.004	14665
q_14179	van	safe	life	guard	1.66	1.26	0.19	0.003	15288
q_14186	stack	seed	maker	hay	2.42	0.68	0.29	0.004	14714
q_14187	start	light	shake	head	1.41	1.42	0.18	0.003	15301
q_14197	slide	mass	lubber	land	2.59	0.73	0.27	0.004	15441

Notes: The values for mean, se (standard error of the mean), and n are based on individual item administrations; the discrimination and difficulty values are based on a 2PL item response model.

Table S7: Item parameters for final CRA items

1.6 Dimensionality and IRT analyses of 30 original CRAT items

NOTE: *The analyses of this sub-section are included here because (a) they are based on the administration of CRAT items (with a mulitple choice response format) to the sample reported on above, and (b) the same IRT procedures used in the previous sub-sections are used again here to identify IRT parameters for these CRAT items for use in Study 2. However, these analyses are not elaborated on extensively in the manuscript as the focus of Study 1 is mainly on the new CRA item set. In Study 4, a more complete pool of CRA items is evaluated together to produce more generalizable IRT parameters, and these are recommended for use by other research teams.*

In addition to the new item set described above (67 original items winnowed down to 65), data were collected on 30 items from an extant CRA item pool (the "CRAT" items; Bowden & Jung-Beeman, 2003) using the revised, multiple-choice response format. These items are traditionally administered with an open-ended response format. In this section, we evaluate the dimensionality and IRT-based item properties of those items as an independent set of 30. This is done as part of validating the new set of items against the older set using the same response choice format and scoring method, though it requires additional evaluation of the psychometric properties of the older set (to support the use of IRT scoring).

In the code chunk below, we evaluated the 30 CRAT items independently; the code is not shown in the PDF as it is duplicative of the methods used earlier with the new item set. As with some of the new items, there was one item among these 30 that had a discrimination near the threshold (item q_14317 = .503); it was dropped. In addition, several of the remaining items had very steep discrimination thresholds, suggesting that conditional (correct/incorrect) response patterns existed among them — this is known as local dependence among the items. To retain as many as possible, we dropped items one-by-one until the model resolved without any items having excessively high discrimination values (this is documented in the .Rnw version of this file). In the end, this left 22 items from the original 30; 1 was dropped due to low discrimination and 7 were dropped to resolve local dependence. The 7 items dropped due to local dependence were:

- q_14307 — **prompt:** way board sleep; **response options:** o,w,h,d,l,s; **answer:** w (walk)
- q_14318 — **prompt:** land hand house; **response options:** a,g,e,l,w,f; **answer:** f (farm)
- q_14310 — **prompt:** fence card master; **response options:** p,t,f,v,l,d; **answer:** p (post)
- q_14308 — **prompt:** tail water flood; **response options:** g,y,r,l,v,s; **answer:** g (gate)
- q_14315 — **prompt:** hold print stool; **response options:** d,l,f,c,b,w; **answer:** f (foot)
- q_14298 — **prompt:** fox man peep; **response options:** r,h,k,m,a,t; **answer:** h (hole)
- q_14299 — **prompt:** shine beam struck; **response options:** r,m,t,c,a,f; **answer:** m (moon)

Note that this does not mean the items above are necessarily "bad" or invalid by themselves (though they may be — more work would be needed to determine this). But, they were removed here because it would not be appropriate to generate IRT-based scores based, in part, on items that are not independent from the remaining items in the set. This also has relevance for Study 2, as described in that sections.

Like the new CRA item set, internal consistency among the retained CRAT items was strong ($\alpha = 0.97$; $\omega_t = 0.97$). Omega general ($\omega_h = 0.92$) also suggests strong unidimensionality in the CRAT set, as shown in Figure S14. Correlations among the items had a mean value of 0.578 and a mean 95% confidence interval range of 0.106. Of the 276 pairwise correlations, 0 had confidence intervals overlapping with zero; 0 of the remaining 276 had point estimates below 0. These estimates are based on pairwise administration counts among the CRAT items of 618.42 on average ($sd = 24.35$; $Mdn = 618$; minimum = 548).

Figures S15 and S13 demonstrate that most of the retained CRAT items are adequately discriminant ($m = 2.29$; $Mdn = 2.04$). The difficulties ($m = 0.68$; $Mdn = 0.7$) are roughly equivalent to those of the new 66 item set. Figure S16 shows the test information and standard error functions; the marginal reliability is 0.85.

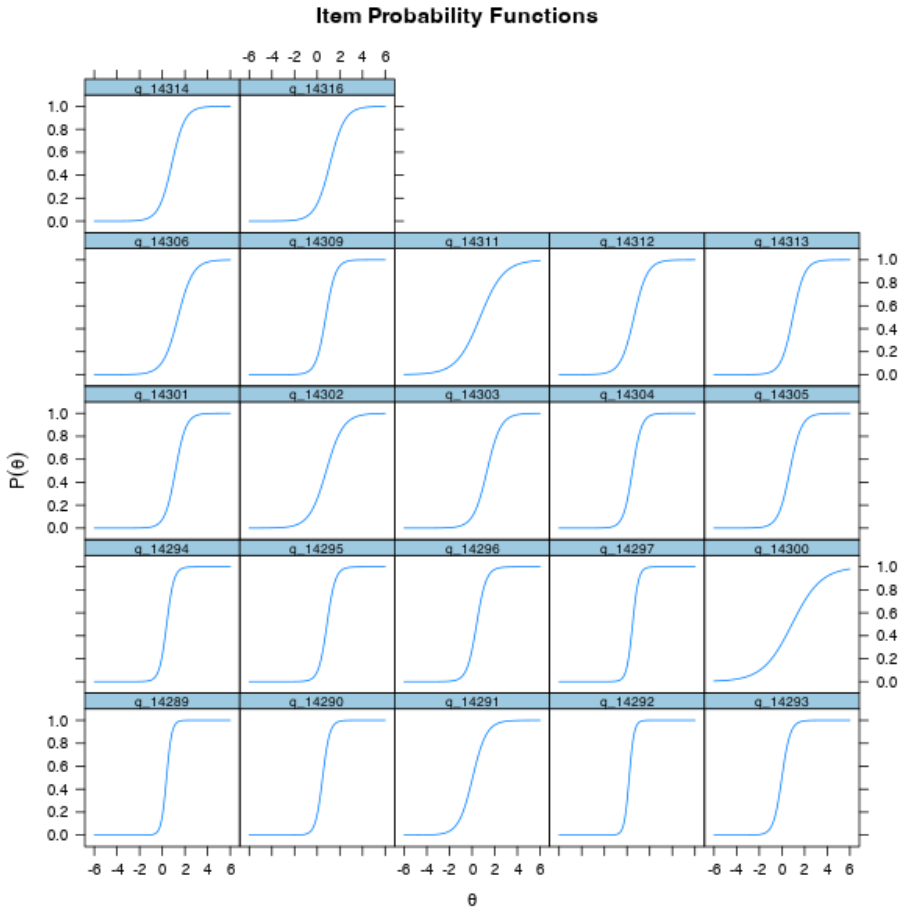


Figure S13: Item probability functions for the retained CRAT items individually.

	Stimuli 1	Stimuli 2	Stimuli 3	Solution	discrimination	difficulty	mean	se	n
q_14289	dew	comb	bee	honey	4.32	0.38	0.37	0.003	23104
q_14290	aid	rubber	wagon	band	3.18	0.48	0.34	0.003	23290
q_14291	flake	mobile	cone	snow	1.62	0.04	0.50	0.003	23171
q_14292	cracker	fly	fighter	fire	4.60	0.19	0.44	0.003	23147
q_14293	cane	daddy	plum	sugar	2.68	-0.00	0.51	0.003	23454
q_14294	dream	break	light	day	3.12	0.39	0.38	0.003	22979
q_14295	fish	mine	rush	gold	2.71	0.87	0.24	0.003	23290
q_14296	worm	shelf	end	book	2.36	0.40	0.38	0.003	23215
q_14297	print	berry	bird	blue	3.80	0.49	0.33	0.003	23101
q_14300	home	sea	bed	sick	0.77	0.89	0.36	0.003	23237
q_14301	sage	paint	hair	brush	2.14	1.16	0.19	0.003	22900
q_14302	mill	tooth	dust	saw	1.33	0.80	0.32	0.003	22994
q_14303	fly	clip	wall	paper	1.66	1.32	0.18	0.003	22929
q_14304	health	taker	less	care	2.48	0.48	0.35	0.003	23147
q_14305	dress	dial	flower	sun	1.93	0.70	0.30	0.003	23045
q_14306	hammer	gear	hunter	head	1.50	1.39	0.18	0.003	22966
q_14309	cover	arm	wear	under	2.30	0.75	0.28	0.003	23311
q_14311	dive	light	rocket	sky	0.96	0.70	0.37	0.003	23020
q_14312	man	glue	star	super	1.69	0.61	0.34	0.003	22974
q_14313	type	ghost	screen	writer	1.91	0.96	0.24	0.003	23053
q_14314	shock	shave	taste	after	1.72	0.85	0.28	0.003	23191
q_14316	pea	shell	chest	nut	1.52	1.13	0.23	0.003	23093

Notes: The values for mean, se (standard error of the mean), and n are based on individual administrations; the discrimination and difficulty values are based on a 2PL item response model.

Table S8: Item parameters for the retained CRAT items

Table S8 includes the coefficients for all item difficulty and discrimination parameters and other descriptive statistics.

These item parameters can be used to IRT score results for these items (when the multiple choice response framework is used) in subsequent studies for the current work.

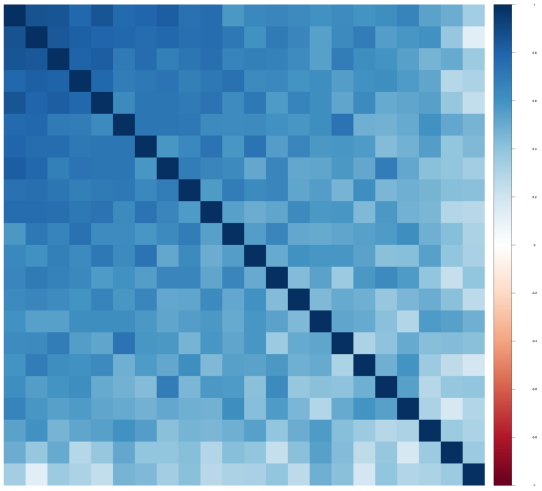


Figure S14: Correlations among the retained CRAT items sorted by general factor loadings.

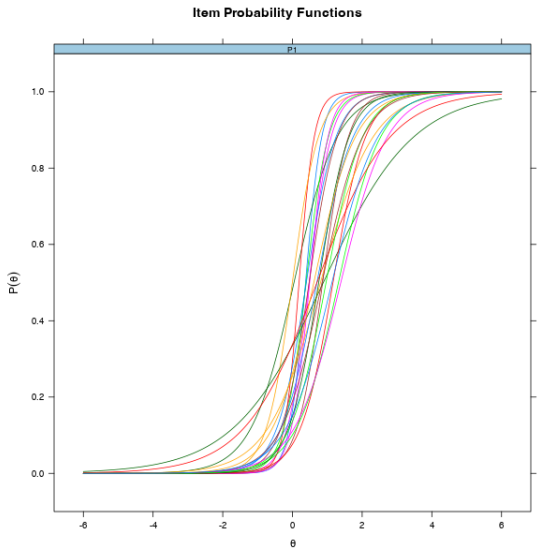


Figure S15: Item probability functions for the retained CRAT items as a bank.

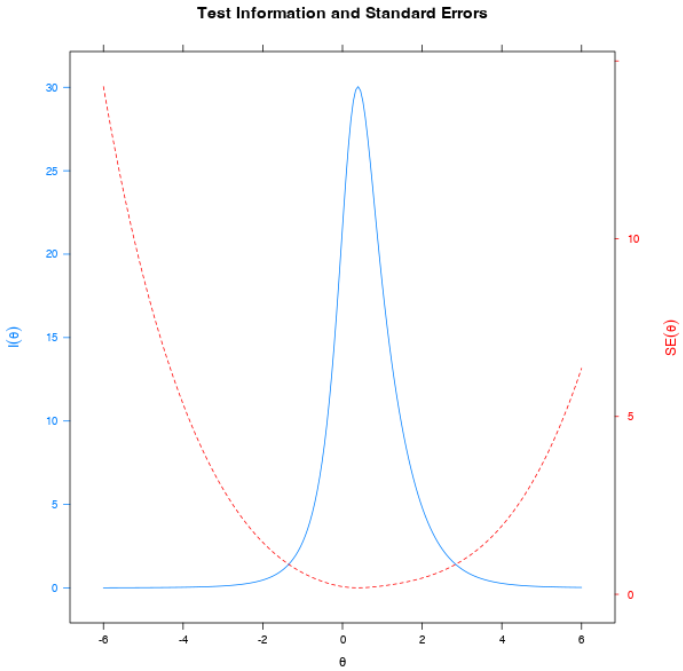


Figure S16: Test information (blue) and standard error (red) functions for the retained CRAT items

1.7 Validation of the final set of new CRA items

In this subsection, we conduct a series of analyses aimed at evaluating the validity of the new CRA item set. Note that the analyses in this sub-section require access to 'version 2' of the data, as the validation data relating to all other variables are not available publicly at the time of submission for peer review. The full dataset will be released at a later date, consistent with prior practice for the [**blinded for review**] project, and all reasonable requests for access prior to this time will be met (contact the senior author to request access).

The first chunk of code for these analyses (not shown in the PDF; see the Rnw version) subsets the main data frame into a more manageable object, one that contains all of the relevant scales scores and demographic variables. The scale scores include IRT-based scores for the 27 traits of the SAPA Personality Inventory (Condon, 2014, 2018) and 6 previously validated scales in the International Cognitive Ability Resource (Blum, Holling, Galibert, & Forthmann, 2016; Condon & Revelle, 2014; Gühne, Doebler, Condon, Luo, & Sun, 2021), as well as scale scores for the Big Five (SPI) using classical test theory (e.g., sum scores). IRT-based scoring procedures were also used for the new CRA item set (65 items) and the CRAT item set (23 items). Descriptive statistics for all of the scales are shown in Table S9.

	n	mean	sd	min	max	range	se
CRA new items	479,070	-0.00	0.57	-1.24	2.47	3.71	0.00
Agreeableness	1,323,449	4.23	0.87	1.00	6.00	5.00	0.00
Conscientiousness	1,323,340	3.96	0.97	1.00	6.00	5.00	0.00
Neuroticism	1,323,392	4.05	1.07	1.00	6.00	5.00	0.00
Extraversion	1,323,355	3.33	1.11	1.00	6.00	5.00	0.00
Openness	1,323,334	4.58	0.80	1.00	6.00	5.00	0.00
Impulsivity	1,107,407	51.47	9.62	28.70	77.14	48.43	0.01
Adaptability	1,109,100	49.99	9.27	22.39	71.14	48.74	0.01
SelfControl	1,103,331	50.69	8.39	24.51	75.83	51.32	0.01
Compassion	1,322,171	50.52	8.68	19.17	66.61	47.44	0.01
Creativity	1,322,099	49.94	8.97	17.25	66.85	49.60	0.01
Charisma	1,114,649	48.87	8.75	24.12	71.75	47.63	0.01
Perfectionism	1,322,177	50.62	8.19	18.35	68.88	50.53	0.01
Conformity	1,112,258	52.21	8.77	27.88	73.15	45.27	0.01
Order	1,321,836	49.84	8.86	24.86	70.94	46.08	0.01
EmotionalStability	1,322,173	48.34	8.87	29.61	75.44	45.83	0.01
Irritability	1,321,563	52.19	9.78	26.94	73.87	46.92	0.01
EmotionalExpressiveness	1,112,461	48.52	9.00	27.21	72.86	45.65	0.01
Intellect	1,322,613	49.30	8.96	17.75	67.58	49.83	0.01
Humor	1,111,129	48.93	8.63	19.86	67.42	47.56	0.01
Sociability	1,322,710	47.41	9.03	26.36	72.66	46.30	0.01
Conservatism	1,100,371	51.12	8.04	32.97	71.05	38.08	0.01
ArtAppreciation	1,109,467	50.53	8.44	24.54	66.84	42.30	0.01
Authoritarianism	1,113,307	50.26	8.18	20.72	70.00	49.28	0.01
Trust	1,321,984	49.03	9.29	23.52	74.02	50.50	0.01
Honesty	1,322,490	49.08	8.86	18.15	66.38	48.22	0.01
AttentionSeeking	1,322,627	50.00	9.82	31.65	76.07	44.42	0.01
WellBeing	1,113,273	47.74	9.52	24.38	67.65	43.26	0.01
SensationSeeking	1,109,251	51.17	9.40	26.78	75.53	48.74	0.01
Introspection	1,321,293	49.59	8.38	17.15	64.54	47.39	0.01
Anxiety	1,322,629	53.23	9.14	25.95	70.42	44.47	0.01
Industry	1,322,001	47.86	8.92	22.16	70.99	48.83	0.01
EasyGoingness	1,104,534	51.67	7.80	21.45	73.21	51.76	0.01
Letter-Number Series	1,179,391	47.82	6.62	34.49	61.20	26.71	0.01
Matrix Reasoning	1,214,525	48.84	5.34	36.39	61.87	25.48	0.00
Rotation3D	1,168,208	49.59	5.92	43.08	72.37	29.29	0.01
Verbal Reasoning	1,218,230	48.00	5.87	29.44	62.01	32.57	0.01
Figural Analogies	893,766	49.70	5.50	38.64	66.24	27.60	0.01
Propositional Reasoning	829,615	49.39	6.21	34.83	67.11	32.28	0.01

Table S9: Descriptive statistics for the new CRA items, SPI-5, SPI-27, and ICAR, scales

Evaluations of scale reliability vary depending on how the scales are scored. For the SPI Big Five, the standardized Cronbach's α values are 0.82 for Agreeableness, 0.82 for Conscientiousness, 0.87 for Neuroticism, 0.86 for Extraversion, and 0.77 for Openness; these are based on the covariances of scores rather than the observed administrations, as is appropriate given the planned missingness design. For the IRT-scored scales, plots of the test information functions and standard errors are shown in Figures S17 to S20.

Once the scales are scored and organized, identifying the point estimates of the correlations and the

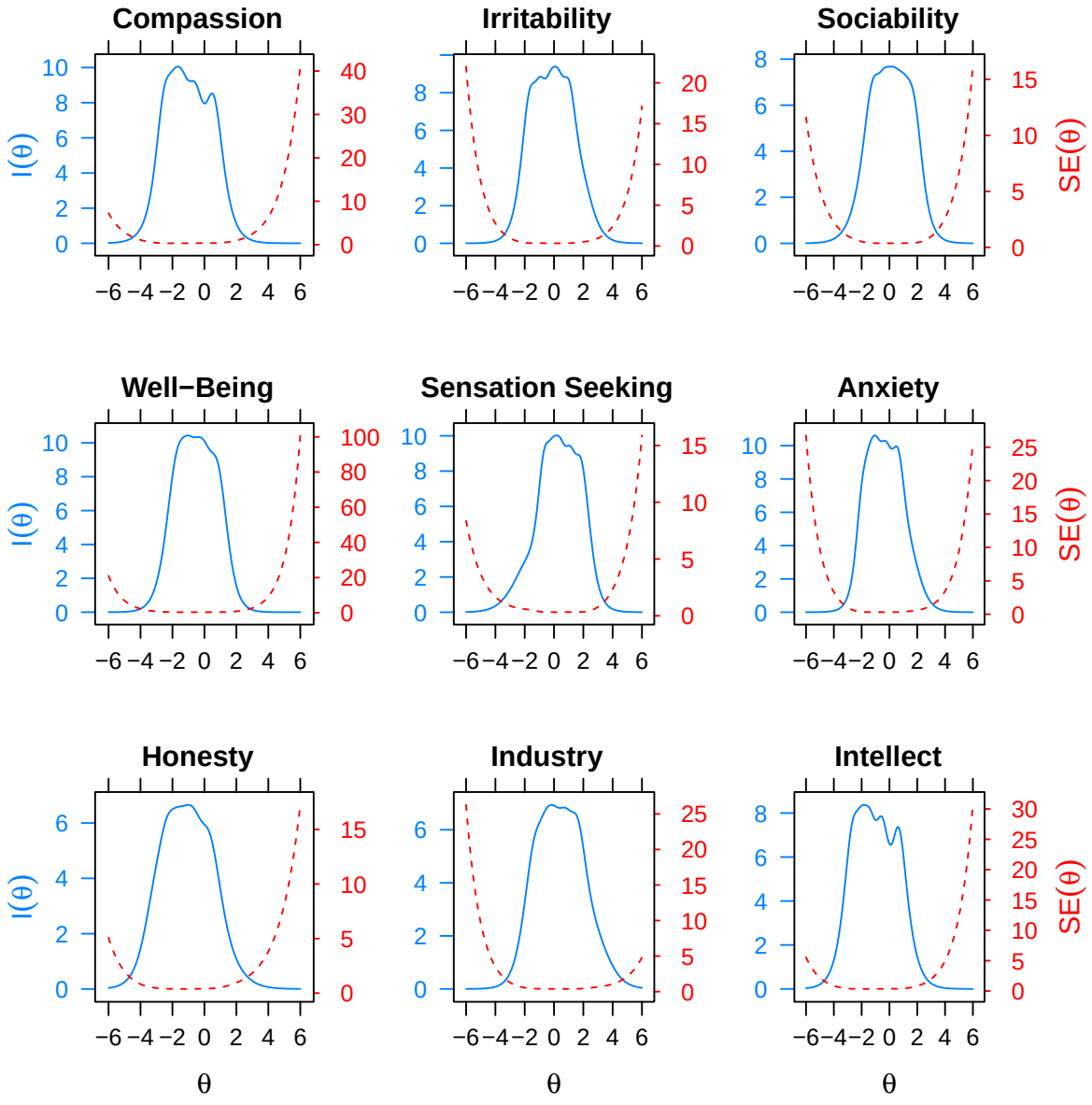


Figure S17: Test information and standard error functions for 9 of the 27 SPI traits

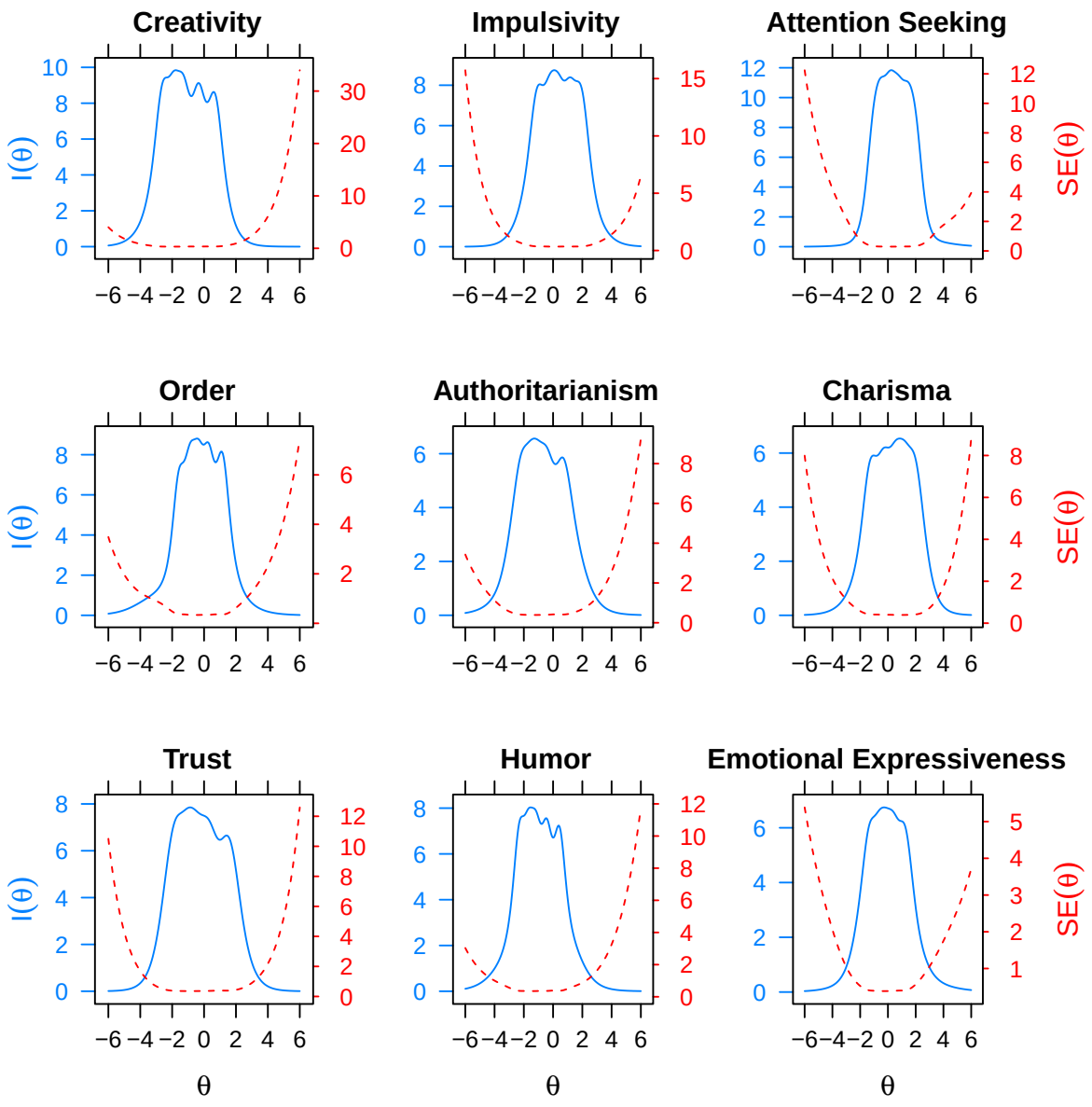


Figure S18: Test information and standard error functions for 9 of the 27 SPI traits

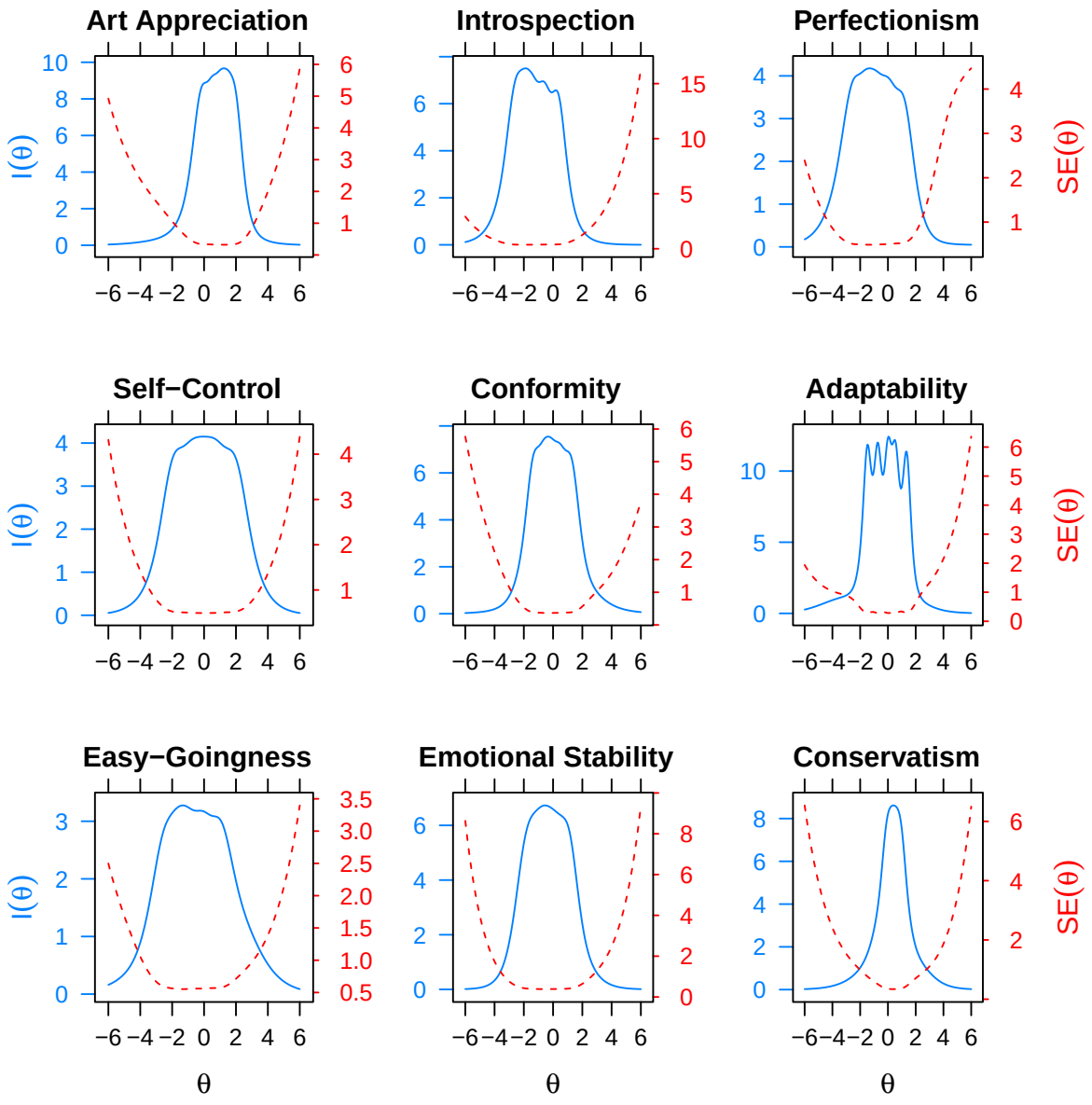


Figure S19: Test information and standard error functions for 9 of the 27 SPI traits

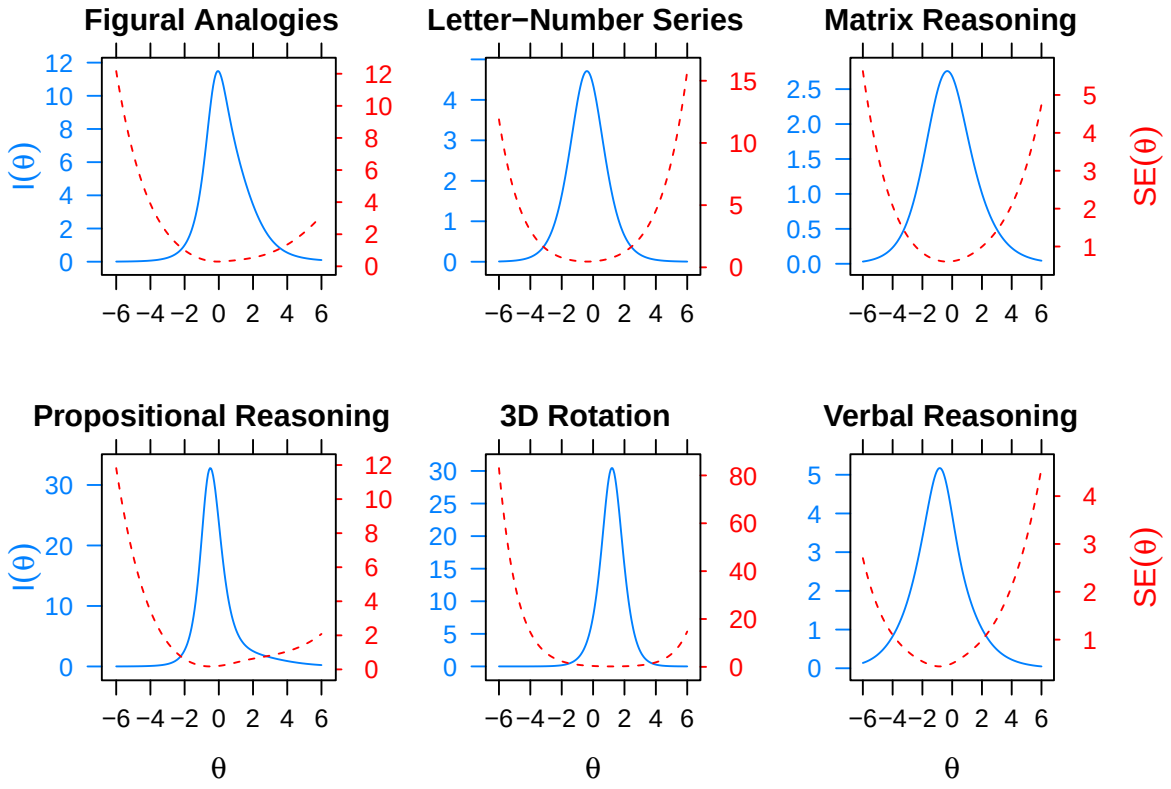


Figure S20: Test information and standard error functions for the ICAR traits

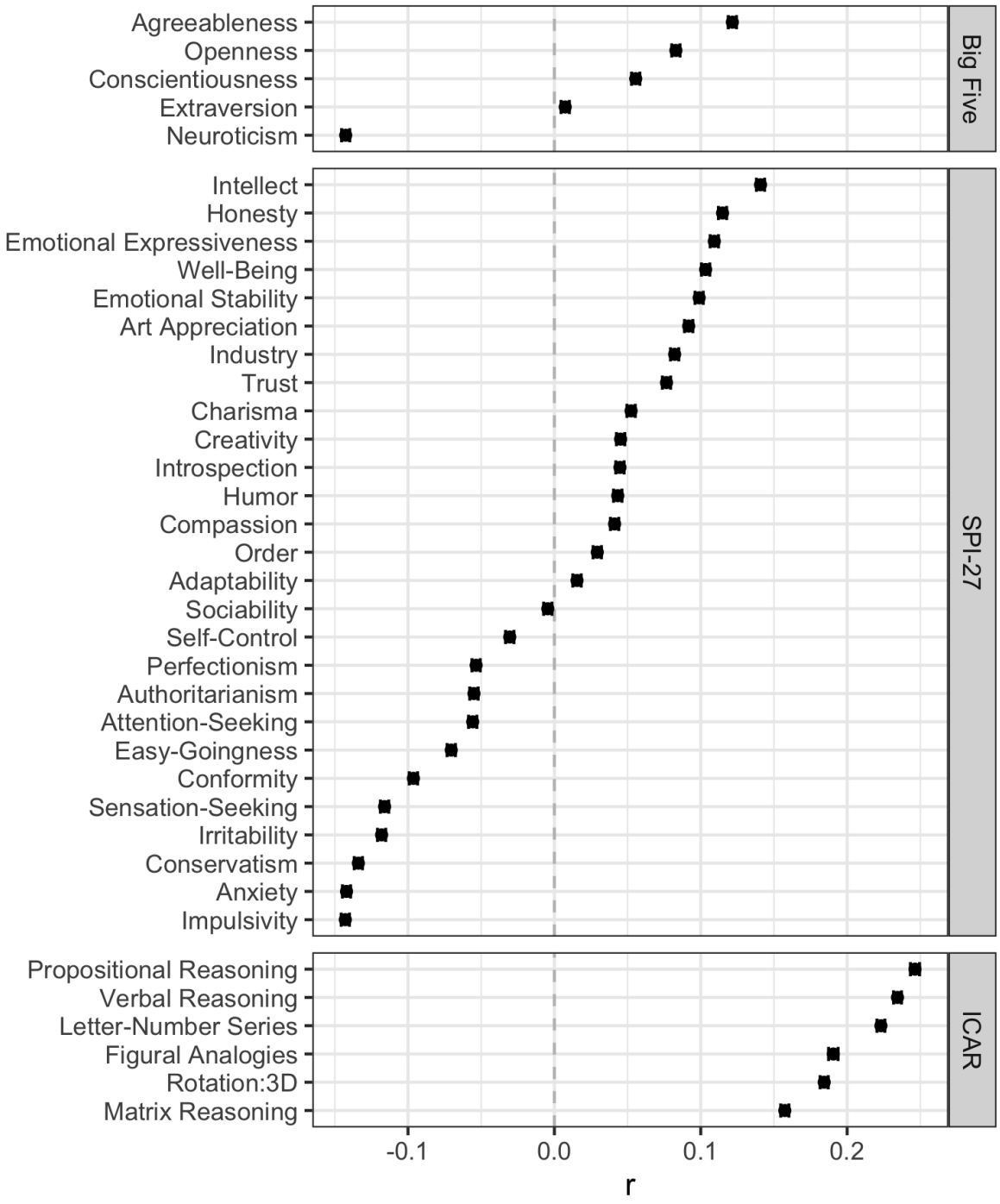


Figure S21: Correlation with the new CRA items

95% confidence intervals is straight-forward. Figure S21 shows the correlations between the new CRA items and the SPI Big Five, SPI-27, and ICAR measures. The results in Figure S21 are mostly consistent with past findings on the relationship between measures of creativity and various other individual differences. As hypothesized, some of the stronger relationships emerged between associative ability and traits related to Openness to Experience. Among the SPI traits, associative ability showed the strongest positive correlation with Intellect, and some of the strongest negative associations with Conservatism (-) and Conformity (-). Unexpectedly, the relationship between associative ability and self-reported Creativity was relatively weak. For the Big Five traits, Openness to Experience showed one of the stronger relationships to associative ability, but the strongest relationship was found with Agreeableness. This was another unexpected finding, as was the relatively strong relationship between associative ability and SPI Honesty, which is closely related to the broader trait of Agreeableness.

In line with previous findings, traits related to emotionality also showed some of the stronger relationships with Associative Ability. For the SPI traits, Associative Ability showed some of the strongest positive relationships with Emotional Expressiveness, Well-Being, and Emotional Stability, and some of the strongest negative relationships with Anxiety, Impulsivity, and Irritability. Associative Ability also showed one of the strongest negative relationships with Big Five Neuroticism.

Associative Ability was more strongly related to the ICAR cognitive abilities than to the temperament traits assessed using the SPI. This could be because the CRA items and cognitive ability measures are both performance-based rather than self-report measures (i.e., the measurement of these constructs introduces shared method variance). Associative Ability showed the strongest relationship with Propositional Reasoning and a relatively strong relationship with Verbal Reasoning. As expected, Associative ability showed weaker relationships with Figural Analogies, Spatial Ability (3D rotation), and Matrix Reasoning. Note that the correlations among the new CRA items and the CRAT items (not shown in the table) were higher than those with the other ICAR item types. The point estimate was 0.49 [95% CI: 0.48,0.51].

	Major	CRA	LN	MR	R3D	VR	FA	PR
1	Classical Languages (Latin and Greek)	0.99	0.84	0.85	0.97	0.99	0.95	0.93
2	American Studies	0.99	0.59	0.75	0.72	0.94	0.63	0.97
3	Poetry Writing	0.98	0.70	0.66	0.68	0.77	0.68	0.75
4	Fiction Writing	0.97	0.66	0.69	0.73	0.86	0.62	0.95
5	Anthropology	0.97	0.80	0.75	0.93	0.97	0.84	0.96
6	Neuroscience	0.96	1.00	1.00	1.00	1.00	1.00	1.00
7	Comparative Literature Studies	0.95	0.77	0.68	0.75	0.84	0.71	0.83
8	Classical Studies	0.94	0.57	0.70	0.92	0.95	0.86	0.92
9	Spanish	0.93	0.64	0.60	0.67	0.97	0.56	0.82
10	Slavic Languages	0.92	0.82	0.87	0.68	0.98	0.77	0.98

Notes: The majors are sorted by percentile on the new CRA item set. LN = Letter-Number Series, MR = Matrix Reasoning, R3D = Three-Dimensional Rotation, VR = Verbal Reasoning, FA = Figural Analogies, PR = Propositional Reasoning.

Table S10: University majors sorted by ICAR percentiles (top 10)

	Occupation	CRA	LN	MR	R3D	VR	FA	PR
1	Education Administrator - Postsecondary	0.96	0.90	0.70	0.86	0.92	0.60	0.84
2	Writer	0.95	0.68	0.54	0.57	0.75	0.66	0.73
3	Physicist	0.94	0.98	0.98	0.99	0.97	0.97	0.97
4	Postsecondary Teacher - Philosophy or Religion	0.94	0.87	0.87	0.78	0.90	0.92	0.93
5	Editor	0.93	0.69	0.66	0.77	0.84	0.71	0.78
6	Environmental Scientist	0.93	0.92	0.87	0.86	0.93	0.87	0.94
7	Veterinarian	0.92	0.81	0.82	0.83	0.79	0.77	0.83
8	Other - Biological Scientist	0.92	0.94	0.96	0.95	0.96	0.94	0.95
9	Postsecondary Teacher - Mathematical Science	0.91	0.93	0.93	0.93	0.88	0.83	0.94
10	Regulatory Affairs Manager	0.91	0.75	0.70	0.62	0.91	0.86	0.95

Notes: The occupations are sorted by percentile on the new CRA item set. LN = Letter-Number Series, MR = Matrix Reasoning, R3D = Three-Dimensional Rotation, VR = Verbal Reasoning, FA = Figural Analogies, PR = Propositional Reasoning.

Table S11: Occupations sorted by ICAR percentiles (top 10)

```
bs_all <- bestScales(CRA_OPR_onet[,c(2:ncol(CRA_OPR_onet))], criteria = cs(CRA_Score),
          cut = .0, n.item = 100, n.iter = 1000,
          overlap = FALSE, check = TRUE, impute = "none", digits = 2)

bs_all_df <- as.data.frame(bs_all$items$CRA_Score)
least <- bs_all_df[order(bs_all_df$mean.r),][1:40,]
most <- bs_all_df[order(-bs_all_df$mean.r),][1:40,]
```

	Freq	mean.r	sd.r
Writing	1000	0.51	0.04
Reading_Comprehension	1000	0.51	0.04
Working_Conditions	1000	0.49	0.04
Recognition	1000	0.49	0.04
Occupational_Prestige	1000	0.48	0.04
Achievement	1000	0.47	0.04
Interpreting_the_Meaning_of_Information_for_Others	1000	0.47	0.04
Critical_Thinking	1000	0.46	0.04
Updating_and_Using_Relevant_Knowledge	1000	0.46	0.04
Inductive_Reasoning	1000	0.46	0.04
Active_Learning	1000	0.46	0.04
Analytical_Thinking	1000	0.45	0.04
Deductive_Reasoning	1000	0.44	0.04
Complex_Problem_Solving	1000	0.44	0.04
Judgment_and_Decision_Making	1000	0.44	0.04
Analyzing_Data_or_Information	1000	0.43	0.04
Independence	1000	0.43	0.04
English_Language	1000	0.42	0.04
Electronic_Mail	1000	0.42	0.04
Systems_Analysis	1000	0.41	0.04
Learning_Strategies	1000	0.41	0.04
Investigative	1000	0.40	0.04
Thinking_Creatively	1000	0.40	0.04
Developing_Objectives_and_Strategies	1000	0.39	0.04
Public_Speaking	1000	0.39	0.04
Fluency_of_Ideas	1000	0.39	0.05
Achievement_Effort	999	0.38	0.04
Category_Flexibility	998	0.36	0.04
Active_Listening	1000	0.36	0.04
Oral_Expression	999	0.36	0.04
Originality	994	0.36	0.05
Innovation	999	0.35	0.04
Persistence	999	0.35	0.04
Instructing	998	0.35	0.05
Initiative	998	0.35	0.04
Speech_Clarity	998	0.35	0.04
Making_Decisions_and_Solving_Problems	993	0.34	0.04
Provide_Consultation_and_Advice_to_Others	994	0.34	0.04
History_and_Archeology	990	0.34	0.04
Operations_Analysis	987	0.34	0.04

Table S12: Job characteristics sorted by association with CRA scores (top 40)

Tables S10 and S11 show the university majors and jobs, respectively, that are most and least associated with scores on the new CRA items among respondents who gave information about their majors and jobs. Both tables are sorted in descending order by CRA percentile, and the tables show ICAR percentile scores for 6 additional cognitive ability domains (Letter-Number Series, Matrix Reasoning, Three Dimensional Rotation, Verbal Reasoning, Figural Analogies, and Propositional Reasoning). For the majors, most of the top 10 were majors that draw heavily on language skills and, presumably, strong vocabularies. The same theme was present for the occupations as well, though there was a slightly broader representation of jobs involving a high general level of education, including 3 university faculty jobs, physicists, and economists in addition to editors, writers, and librarians.

Tables S12 and S13 show the O*NET job characteristics that are most and least, respectively, associated with scores on the new CRA items among respondents who gave information about their jobs. As described in the main manuscript, the job characteristics are collected by O*NET directly; the analyses are based on the BISCUIT technique (Elleman, McDougald, Condon, & Revelle, 2020) using 1,000 bootstrapped iterations. Here again, the job characteristics most highly associated with CRA scores include strong language skills, especially reading and writing. The job characteristics least associated with CRA scores are entirely

	Freq	mean.r	sd.r
Spend_Time_Walking_and_Running	1000	-0.41	0.04
Spend_Time_Bending_or_Twisting_the_Body	1000	-0.39	0.04
Spend_Time_Standing	999	-0.34	0.04
Extent_Flexibility	1000	-0.34	0.04
Handling_and_Moving_Objects	999	-0.34	0.04
Trunk_Strength	1000	-0.34	0.04
Performing_General_Physical_Activities	999	-0.34	0.04
Speed_of_Limb_Movement	1000	-0.33	0.04
Stamina	999	-0.33	0.04
Static_Strength	997	-0.32	0.04
Dynamic_Strength	996	-0.32	0.04
Manual_Dexterity	998	-0.32	0.04
Multilimb_Coordination	991	-0.32	0.04
Gross_Body_Coordination	991	-0.31	0.04
Food_Production	935	-0.31	0.04
Realistic	959	-0.30	0.04
Exposed_to_Minor_Burns_Cuts_Bites_or_Stings	955	-0.30	0.04
Arm-Hand_Steadiness	954	-0.30	0.04
Spend_Time_Kneeling_Crouching_Stooping_or_Crawling	967	-0.30	0.04
Pace_Determined_by_Speed_of_Equipment	955	-0.29	0.04
Spend_Time_Using_Your_Hands_to_Handle_Control_or_Feel_Objects_Tools_or_Controls	922	-0.29	0.04
Spend_Time_Keeping_or_Regaining_Balance	904	-0.29	0.04
Very_Hot_or_Cold_Temperatures	918	-0.29	0.04
Philosophy_and_Theology	909	0.30	0.04
Information_Ordering	920	0.31	0.04
Computers_and_Electronics	951	0.31	0.04
Communications_and_Media	969	0.31	0.04
Getting_Information	978	0.32	0.04
Memorization	980	0.32	0.04
Science	967	0.33	0.04
Structured_versus_Unstructured_Work	977	0.33	0.04
Indoors_Environmentally_Controlled	994	0.33	0.04
Oral_Comprehension	992	0.33	0.04
Freedom_to_Make_Decisions	980	0.33	0.04
Organizing_Planning_and_Prioritizing_Work	986	0.33	0.04
Processing_Information	986	0.33	0.04
Spend_Time_Sitting	995	0.33	0.04
Operations_Analysis	987	0.34	0.04
History_and_Archeology	990	0.34	0.04
Provide_Consultation_and_Advice_to_Others	994	0.34	0.04

Table S13: Job characteristics sorted by association with CRA scores (bottom 40)

represented by physical tasks unrelated to communication and language.

2 Study 2

The aim of Study 2 was to validate the new CRA item set from Study 1 in a student sample relative to academic achievement measures, and to evaluate the effects of response format (open-ended vs multiple-choice) and item type (CRAT vs new CRA items). Data were collected from an undergraduate sample at **[blinded for review]**, starting on September 30, 2020 and ending on March 21, 2021. These data are available on OSF (tinyurl.com/2a8vx6n8) in a file labeled 'Study2data.csv'. In the subsections below, we describe the steps for loading and cleaning the data, and carrying out the planned analyses.

2.1 Data wrangling

Begin by loading, cleaning, and scoring the data. Note that the data have already been de-identified, and that responses to the individual items have already been scored as correct (1) or incorrect (0). Half of the items were given using an open-response format, and these were coded manually by the first author. As the respondents were undergraduates participating for course credit, they were given the option of not responding. Items that were skipped or marked as "I don't know" were scored as incorrect.

To score the items, two approaches are used. For the open-ended response format, we used sum/mean scores, scoring separately for the new CRA item content and for the existing CRAT content. For the multiple choice response format, we used sum/mean scores and IRT scoring based on the item calibrations identified

in Study 1. This meant that some items were dropped due to psychometric properties, as described in Study 1. Note also that two of the items administered in Study 1 were missing from this data set: item q_14012 was omitted from data collection due to a coding error and item q_14125 was used as an example item in this data collection.

```
rm(list=ls()[!(ls() %in% c('findcolnumber', 'keys.list', 'wdpath', 'score_ICAR_IRT', 'CRAitemNumbersStudy1'))])

# Load the de-identified data file. The raw data file has been cleaned to
# remove identifying data that is automatically collected by Qualtrics.
study2data <- read.csv(paste(wdpath, "data/study2/study2data.csv", sep=""), header=TRUE)

# Drop participants who self-report low English-language fluency
study2data <- study2data[study2data$English_Fluency != "Not well", ]

# Get the sum scores for the open-ended response format for the new CRA item content
study2data$meanScore_CRAopen <- rowMeans(study2data[,c(findcolnumber(study2data, "q_14003_CRA_open1scored"):
                                                       findcolnumber(study2data, "q_14197_CRA_open65scored"))],
                                                       na.rm = TRUE)
# Get the sum scores for the open-ended response format for the older CRAT items
study2data$meanScore_CRATopen <- rowMeans(study2data[,c(findcolnumber(study2data, "q_14289_CRAT_open1scored"):
                                                       findcolnumber(study2data, "q_14318_CRAT_open30scored"))],
                                                       na.rm = TRUE)
# Get the sum scores for the multiple choice response format for the new CRA item content
study2data$meanScore_CRA_MC <- rowMeans(study2data[,c(findcolnumber(study2data, "q_14003_CRA_MC1scored"):
                                                       findcolnumber(study2data, "q_14197_CRA_MC65scored"))],
                                                       na.rm = TRUE)
# Get the sum scores for the multiple choice response format for the older CRAT items
study2data$meanScore_CRAT_MC <- rowMeans(study2data[,c(findcolnumber(study2data, "q_14289_CRAT_MC1scored"):
                                                       findcolnumber(study2data, "q_14318_CRAT_MC30scored"))],
                                                       na.rm = TRUE)

# In addition to sum scoring, we also use IRT scoring for the MC version of the CRA and CRAT items
# The calibrations for the IRT scores are based on the results of Study 1.

# To IRT score the new CRA item set, two of the items need to be dropped based on psychometric
# evaluation in Study 1 (items 14043 and 14023). For the CRAT items, an additional 7 items also
# should be left out.
MCitemsForIRT <- study2data[,-c(findcolnumber(study2data, "q_14023_CRA_MC11scored"),
                                findcolnumber(study2data, "q_14043_CRA_MC15scored"),
                                findcolnumber(study2data, "q_14307_CRAT_MC19scored"),
                                findcolnumber(study2data, "q_14308_CRAT_MC20scored"),
                                findcolnumber(study2data, "q_14310_CRAT_MC22scored"),
                                findcolnumber(study2data, "q_14315_CRAT_MC27scored"),
                                findcolnumber(study2data, "q_14317_CRAT_MC29scored"),
                                findcolnumber(study2data, "q_14318_CRAT_MC30scored"),
                                findcolnumber(study2data, "q_14298_CRAT_MC30scored")
                                )]

# Isolate the new CRA items to IRT score
CRAitemSetFinal <- MCitemsForIRT[,c(findcolnumber(MCitemsForIRT, "q_14003_CRA_MC1scored"):
                                    findcolnumber(MCitemsForIRT, "q_14197_CRA_MC65scored"))]
# Align the item labels
names(CRAitemSetFinal) <- substring(names(CRAitemSetFinal), 1, 7)

# Now load the IRT model and IRT score
load(file = paste(wdpath, "data/study1/IRTmodelCRAnewItems.rdata", sep = ""))
CRAitemSetFinal = score_ICAR_IRT(dataset = CRAitemSetFinal, IRT_model = IRT_model_ICAR_CRAnew, t_score = FALSE)

# Isolate the CRAT items to IRT score
CRATitemSetFinal <- MCitemsForIRT[,c(findcolnumber(MCitemsForIRT, "q_14289_CRAT_MC1scored"):
                                    findcolnumber(MCitemsForIRT, "q_14316_CRAT_MC28scored"))]
# Align the item labels
names(CRATitemSetFinal) <- substring(names(CRATitemSetFinal), 1, 7)

# Now load the IRT model and IRT score
load(file = paste(wdpath, "data/study1/IRTmodelCRATitems.rdata", sep = ""))
CRATitemSetFinal = score_ICAR_IRT(dataset = CRATitemSetFinal, IRT_model = IRT_model_CRAT, t_score = FALSE)

# Tack the IRT scores onto the full data frame for Study 2
study2data <- data.frame(study2data, CRAitemSetFinal$ICAR_CRA, CRAitemSetFinal$SE_ICAR_CRA,
                         CRATitemSetFinal$CRAT, CRATitemSetFinal$SE_CRAT)

names(study2data)[names(study2data) == 'CRAitemSetFinal.ICAR_CRA'] <- 'IRTscore_CRA_MC'
names(study2data)[names(study2data) == 'CRAitemSetFinal.SE_ICAR_CRA'] <- 'IRTscore_CRA_MC_SE'
names(study2data)[names(study2data) == 'CRATitemSetFinal.CRAT'] <- 'IRTscore_CRAT_MC'
names(study2data)[names(study2data) == 'CRATitemSetFinal.SE_CRAT'] <- 'IRTscore_CRAT_MC_SE'
```

```
# Save objects out for later use
study2dataCRAonly <- subset(CRAitemSetFinal, select = c(q_14003:q_14197))
CRAitemNumbersStudy2 <- colnames(study2dataCRAonly)
save(study2dataCRAonly, CRAitemNumbersStudy2, file = paste(wdpath, "data/study2/CRAdataStudy2.rdata", sep = ""))

# Compare the two methods for scoring the multiple choice responses
compareScoringMethodCor <- round(cor(study2data[,c("meanScore_CRA_MC", "meanScore_CRAT_MC",
                                "IRTscore_CRA_MC", "IRTscore_CRAT_MC")], use="pairwise"), 2)
compareScoringMethodCor
```

```
##                   meanScore_CRA_MC meanScore_CRAT_MC IRTscore_CRA_MC IRTscore_CRAT_MC
## meanScore_CRA_MC              1.00              0.45            0.93             0.45
## meanScore_CRAT_MC             0.45              1.00            0.47             0.89
## IRTscore_CRA_MC               0.93              0.47            1.00             0.48
## IRTscore_CRAT_MC              0.45              0.89            0.48             1.00
```

Results suggest that the two methods for scoring the multiple choice items are similar but not identical. The associations among the mean and IRT scores for the new CRA items is 0.93, and 0.89 for the CRAT items. Though high, the correlations would likely be higher still if the items were more fully overlapping (recall that some items were omitted from both sets, mainly due to the removal of items with relatively poor psychometric properties when calculating the IRT scores). The correlations between the new CRA items and the CRAT items are similar regardless of scoring method (0.45 for mean scores and 0.48 for IRT scores). Given the availability of the IRT calibrations, we recommend this method and will use these scores for all subsequent analyses in Studies 2 and 3.

2.2 Descriptive statistics

Now that the data have been wrangled into shape, the next code chunk describes the sample (N = 684) and key variables before moving on to the validation analyses.

```
# Scrub the data to remove implausible/impossible values for self reports of test scores and GPA
study2data <- psych::scrub(study2data, where = "ACT_Composite_Score", min = 1, max = 36)
study2data <- psych::scrub(study2data, where = "SAT_RW_Score", min = 400, max = 800)
study2data <- psych::scrub(study2data, where = "SAT_M_Score", min = 400, max = 800)
study2data <- psych::scrub(study2data, where = "GPA", min = 0, max = 5)
study2data$Ethnicity <- str_replace(study2data$Ethnicity, "Some other ethnicity or origin, please specify:", "Other")
study2data$Ethnicity <- str_replace(study2data$Ethnicity, "I prefer not to answer.", "Prefer not to say")

descriptives <- describe(study2data)

ESL <- data.frame(table(study2data$English_First_Language)[2:3])
colnames(ESL) <- c("English 1st Language?", "Participants")
ESL$Percentage <- round(ESL$Participants/sum(ESL$Participants)*100,2)

english <- data.frame(table(study2data$English_Fluency)[c(2,1)])
colnames(english) <- c("English fluency", "Participants")
english$Percentage <- round(english$Participants/sum(english$Participants)*100,2)

edu <- data.frame(table(study2data$Education)[c(4,3,5,1,2)])
colnames(edu) <- c("Educational Attainment Level", "Participants")
edu$Percentage <- round(edu$Participants/sum(edu$Participants)*100,2)

income <- data.frame(table(study2data$Income)[c(12,5,9,10,11,2,3,4,6,7,8,13,14)])
colnames(income) <- c("Household Income", "Participants")
income$Percentage <- round(income$Participants/sum(income$Participants)*100,2)

gender <- data.frame(table(study2data$Gender_Identity))
colnames(gender) <- c("Gender Identity", "Participants")
gender$Percentage <- round(gender$Participants/sum(gender$Participants)*100,2)

ethnicity <- data.frame(table(study2data$Ethnicity)[c(1:6,9,7,8)])
colnames(ethnicity) <- c("Race/Ethnicity", "Participants")
ethnicity$Percentage <- round(ethnicity$Participants/sum(ethnicity$Participants)*100,2)

descriptives[c("Age", "Duration_Minutes", "SAT_RW_Score", "SAT_M_Score", "ACT_Composite_Score", "GPA",
               "meanScore_CRAopen", "meanScore_CRATopen", "IRTscore_CRA_MC", "IRTscore_CRA_MC_SE",
               "IRTscore_CRAT_MC", "IRTscore_CRAT_MC_SE"),c("n","mean","sd", "median", "min", "max")]
```

```
##                       n   mean     sd median    min     max
```

```
## Age                 684  19.54   2.36  19.00  18.00   47.00
## Duration_Minutes    684 105.84 617.06  27.29   5.17 8320.90
## SAT_RW_Score         98 650.64  74.44 650.00 400.00  790.00
## SAT_M_Score         118 627.94  75.55 620.00 410.00  800.00
## ACT_Composite_Score 213  26.64   3.98  27.00  17.00   35.00
## GPA                 452   3.48   0.45   3.50   1.60    4.30
## meanScore_CRAopen   684   0.10   0.09   0.10   0.00    0.45
## meanScore_CRATopen  675   0.24   0.19   0.20   0.00    0.80
## IRTscore_CRA_MC     675  -0.11   0.65  -0.03  -1.37    2.13
## IRTscore_CRA_MC_SE  675   0.40   0.11   0.36   0.24    0.86
## IRTscore_CRAT_MC    673   0.06   0.71   0.11  -1.05    1.70
## IRTscore_CRAT_MC_SE 673   0.58   0.13   0.58   0.26    0.94
```

Descriptive statistics for (most of) the continuous variables (i.e., age, test scores, CRA scores) are described as output in the chunk above. Note that these include the standard errors for the IRT scores of the new CRA scale ($m = 0.4$) and the CRAT scale ($m = 0.58$). Categorical variables are described in Tables S14 to S19.

English 1st Language?	Participants	Percentage
No	47	6.88
Yes	636	93.12

Table S14: English as a first language in the Study 2 sample

English fluency	Participants	Percentage
Well	30	4.39
Very well (fluent/native)	654	95.61

Table S15: English fluency in the Study 2 sample

Educational attainment level	Participants	Percentage
High school graduate/GED	12	1.75%
Currently in college/university	621	90.79%
Some college/university, but did not graduate	17	2.49%
Associate degree	10	1.46%
College/university degree	24	3.51%

Table S16: Educational attainment in the Study 2 sample

2.3 Correlations among CRA scores, GPA, and standardized test scores

The preregistration specified that correlations would be reported (in the main manuscript) if there were at least 200 pairwise observations. The number of observations was not above this threshold for the SAT verbal (98) or SAT math (118) scores, but the ACT composite scores (N = 213) and GPAs were more than adequate (N = 452). These analyses are run in the following code chunk and shown at the end.

```
# Subset the relevant scores from the main data set
scoresSubset <- subset(study2data, select = c(SAT_RW_Score, SAT_M_Score, ACT_Composite_Score,
                                              GPA, meanScore_CRAopen, meanScore_CRATopen,
                                              IRTscore_CRA_MC, IRTscore_CRAT_MC))
scoresCor <- cor.ci(scoresSubset, plot = FALSE)

# But only some of these should be reported given the preregistration
```

Household income (annual)	Participants	Percentage
Less than 20,000	124	18.18%
20,000 to 40,000	66	9.68%
40,000 to 60,000	40	5.87%
60,000 to 80,000	51	7.48%
80,000 to 100,000	40	5.87%
100,000 to 120,000	52	7.62%
120,000 to 150,000	35	5.13%
150,000 to 200,000	33	4.84%
200,000 to 250,000	35	5.13%
250,000 to 350,000	20	2.93%
350,000 to 500,000	23	3.37%
More than 500,000	17	2.49%
Prefer not to say	146	21.41%

Table S17: Household income in the Study 2 sample

Gender identity	Participants	Percentage
Female	464	67.84%
Male	200	29.24%
Non-binary	15	2.19%
Prefer not to say	4	0.58%
Prefer to self-describe	1	0.15%

Table S18: Gender identity in the Study 2 sample

Race/ethnicity	Participants	Percentage
American Indian or Alaska Native	5	0.73%
Asian	79	11.55%
Black or African American	30	4.39%
Hispanic, Latinx or Spanish Origin	85	12.43%
Middle Eastern or North African	5	0.73%
Native Hawaiian or Other Pacific Islander	4	0.58%
White	443	64.77%
Other	22	3.22%
Prefer not to say	11	1.61%

Table S19: Race/ethnicity in the Study 2 sample

```r
scoresSubset2 <- subset(study2data, select = c(ACT_Composite_Score, GPA,
                                               meanScore_CRAopen, meanScore_CRATopen,
                                              IRTscore_CRA_MC, IRTscore_CRAT_MC))
scoresCor2 <- cor.ci(scoresSubset2, plot = FALSE)

# Get the correlations
cors <- round(scoresCor2$rho, 2)
# and the 95% confidence intervals
CIs <- round(scoresCor2$ci[,c("lower", "upper", "p")], 3)
cors
```

```
##                     ACT_Composite_Score  GPA meanScore_CRAopen meanScore_CRATopen IRTscore_CRA_MC IRTscore_CRAT_MC
## ACT_Composite_Score                1.00 0.25              0.33               0.32            0.19             0.18
## GPA                                0.25 1.00              0.14               0.04            0.17             0.15
## meanScore_CRAopen                  0.33 0.14              1.00               0.44            0.36             0.26
## meanScore_CRATopen                 0.32 0.04              0.44               1.00            0.38             0.43
## IRTscore_CRA_MC                    0.19 0.17              0.36               0.38            1.00             0.48
## IRTscore_CRAT_MC                   0.18 0.15              0.26               0.43            0.48             1.00
```

```r
CIs
```

```
##                    lower upper     p
## ACT_C-GPA          0.092 0.388 0.002
## ACT_C-mnS_CRA      0.239 0.432 0.000
## ACT_C-mS_CRAT      0.198 0.438 0.000
## ACT_C-IRT_CRA_     0.064 0.321 0.004
## ACT_C-IRT_CRAT     0.067 0.305 0.003
## GPA-mnS_CRA        0.052 0.245 0.003
## GPA-mS_CRAT       -0.048 0.148 0.312
## GPA-IRT_CRA_       0.069 0.275 0.001
## GPA-IRT_CRAT       0.059 0.239 0.001
## mnS_CRA-mS_CRAT    0.385 0.496 0.000
## mnS_CRA-IRT_CRA_   0.290 0.432 0.000
## mnS_CRA-IRT_CRAT   0.184 0.340 0.000
## mS_CRAT-IRT_CRA_   0.317 0.447 0.000
## mS_CRAT-IRT_CRAT   0.368 0.495 0.000
## IRT_CRA_-IRT_CRAT  0.433 0.527 0.000
```

ACT composite score was significantly correlated with both CRA and CRAT item scores across response modes (i.e., multiple-choice and open-ended), but correlations were stronger when using the open-ended response mode. GPA showed significant correlations with CRA item scores when using both multiple-choice and open-ended response mode. The correlation between GPA and CRAT item scores was significant when using a multiple-choice response mode but not when using an open-ended response mode. The difference in the magnitudes of the correlations across response choices for the ACT only (i.e., not GPA) may have reflected the greater difficulty of the open-ended response method (see below), as there is considerable evidence that the ACT and other standardized tests are more direct measures of cognitive ability than GPA (Galla et al., 2019). GPA, by contrast, is also affected by other aspects of personality (e.g., self-regulation skills).

All correlations between the same measure when using different response modes (e.g., multiple-choice CRA items and open-ended CRA items), and different measures using the same response mode (e.g., open-ended CRA items and open-ended CRAT items) were significant. The strongest relationship across combinations of measure (CRA and CRAT) and response mode (multiple-choice and open-ended) emerged between the CRA and CRAT items using the multiple-choice format. This could be due to both the relative ease of solving the multiple-choice items and the greater precision afforded by IRT scoring, which was not possible for the open-ended items.

Note that exploratory analyses of the correlations with SAT verbal scores were statistically significant for the new CRA items with the open-ended response format (r=.25, p=.01) and for the IRT-scored multiple-choice format (r=.29, p=.006), despite the small sample size. None of the other SAT correlations were statistically significant (p<.05).

2.4 Effects of response format and problem set

In order to evaluate the effect of response format (open-ended vs multiple choice) and of problem set (CRAT vs new CRA items), it is first necessary to pivot the data into long format (i.e., one row for each

item for each participant). This is done below, along with dummy coding according to response and problem type.

```
# Getting the data into long form
Study2_Data_Long <- study2data %>%
                    pivot_longer(cols = c(q_14003_CRA_open1scored:q_14318_CRAT_MC30scored),
                    names_to = "Item",
                    values_to = "Score")
# Creating dummy variables for problem set and response mode
Study2_Data_Long <- Study2_Data_Long %>%
                    mutate(Set = ifelse(str_detect(Item, "CRAT"), 0, 1),
                    Format = ifelse(str_detect(Item, "open"), 0, 1))
# Problem set and response mode variables as factors
Study2_Data_Long$Set <- factor(Study2_Data_Long$Set,
                            levels = c(0,1),
                            labels = c("CRAT","CRA"))
Study2_Data_Long$Format <- factor(Study2_Data_Long$Format,
                            levels = c(0,1),
                            labels = c("Open-ended","Multiple-choice"))
```

Then, we begin by modelling just the main effects of (1) problem set (CRAT vs CRA) and (2) response format (open-ended vs multiple-choice). This is done in the code chunk below, using a binary logitistic multilevel model with item scores predicted by 'set' and 'format'.

```
Format_Model_main <- glmer(Score ~ Format + Set + (1 | ResponseId), data = Study2_Data_Long, family = "binomial")
summary(Format_Model_main)

## Generalized linear mixed model fit by maximum likelihood (Laplace Approximation) ['glmerMod']
##  Family: binomial  ( logit )
## Formula: Score ~ Format + Set + (1 | ResponseId)
##    Data: Study2_Data_Long
##
##      AIC      BIC   logLik deviance df.resid
##  31950.2  31984.2 -15971.1  31942.2    36788
##
## Scaled residuals:
##     Min      1Q  Median      3Q     Max
## -1.5840 -0.4782 -0.3597 -0.2377  5.5014
##
## Random effects:
##  Groups     Name        Variance Std.Dev.
##  ResponseId (Intercept) 0.5      0.7071
## Number of obs: 36792, groups:  ResponseId, 684
##
## Fixed effects:
##                       Estimate Std. Error z value Pr(>|z|)
## (Intercept)           -1.41111    0.03835  -36.80   <2e-16 ***
## FormatMultiple-choice  0.64107    0.02927   21.91   <2e-16 ***
## SetCRA                -0.88703    0.03047  -29.11   <2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Correlation of Fixed Effects:
##             (Intr) FrmtM-
## FrmtMltpl-c -0.302
## SetCRA      -0.411 -0.208
```

	Estimate	CI lower	CI upper	Odds Ratio
Intercept	-1.41	-1.49	-1.34	0.24
Response Format (Open vs MC)	0.64	0.58	0.70	1.90
Problem Set (CRAT v CRA)	-0.89	-0.95	-0.83	0.41

Notes: The estimate values are the logistic regression coefficients. CI lower and upper represent the bounds of the 95 confidence interval.
Reference group is open-ended CRAT items.

Table S20: Model summary testing the main effects

A summary of the results are shown in the code and output above, as well as in Table S20. The odds ratio values indicate that the change in odds of correctly responding to the items increases by 1.9 from CRAT items with an open-ended response format to CRAT items with a multiple-choice format. The odds decrease

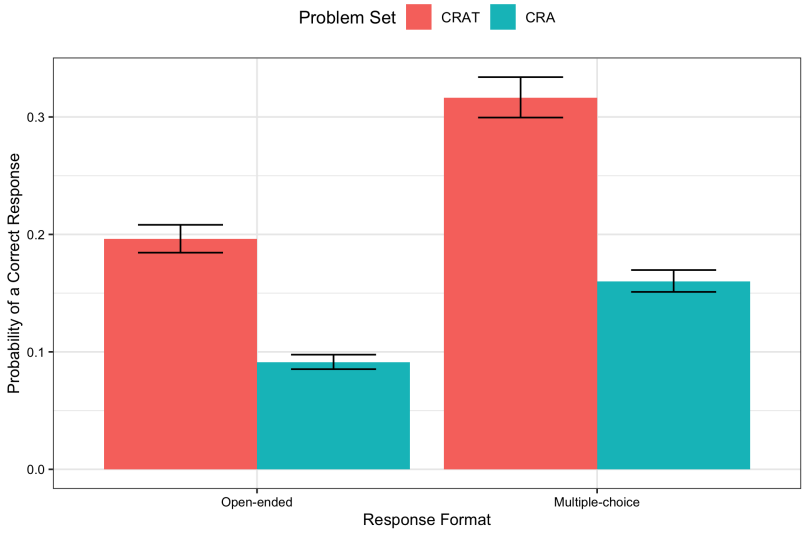


Figure S22: Effects of response format and item type on CRA scores

when going from CRAT items to the new CRA items (odds are 0.41). These results are also reflected in Figure S22.

To follow up on the main effects analysis, we then identify a model that also includes the interaction effect between problem format and problem set. The next code chunk shows the results.

```
Format_Model_interaction <- glmer(Score ~ Format*Set + (1 | ResponseId), data = Study2_Data_Long, family = "binomial")
summary(Format_Model_interaction)

## Generalized linear mixed model fit by maximum likelihood (Laplace Approximation) ['glmerMod']
##  Family: binomial  ( logit )
## Formula: Score ~ Format * Set + (1 | ResponseId)
##    Data: Study2_Data_Long
##
##      AIC      BIC   logLik deviance df.resid
##  31869.3  31911.8 -15929.6  31859.3    36787
##
## Scaled residuals:
##     Min      1Q  Median      3Q     Max
## -1.4196 -0.4884 -0.3581 -0.2350  5.8545
##
## Random effects:
##  Groups     Name        Variance Std.Dev.
##  ResponseId (Intercept) 0.4997   0.7069
## Number of obs: 36792, groups:  ResponseId, 684
##
## Fixed effects:
##                             Estimate Std. Error z value Pr(>|z|)
## (Intercept)                 -1.27642    0.04054 -31.484  < 2e-16 ***
## FormatMultiple-choice        0.28254    0.04923   5.739 9.54e-09 ***
## SetCRA                      -1.14687    0.04184 -27.414  < 2e-16 ***
## FormatMultiple-choice:SetCRA  0.56267    0.06172   9.117  < 2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Correlation of Fixed Effects:
##             (Intr) FrmtM- SetCRA
## FrmtMltpl-c -0.437
## SetCRA      -0.504  0.419
## FrmtM-:SCRA  0.346 -0.797 -0.676
```

In this case, the results are similar (directionally) and there is a statistically significant effect of the

	Estimate	CI lower	CI upper	Odds Ratio
(Intercept)	-1.28	-1.36	-1.20	0.28
Response Format (Open vs MC)	0.28	0.19	0.38	1.33
Problem Set (CRAT v CRA)	-1.15	-1.23	-1.06	0.32
Interaction	0.56	0.44	0.68	1.76

Notes: The estimate values are the logistic regression coefficients. CI lower and upper represent the bounds of the 95 confidence interval.

Table S21: Model summary testing the main effects and interaction

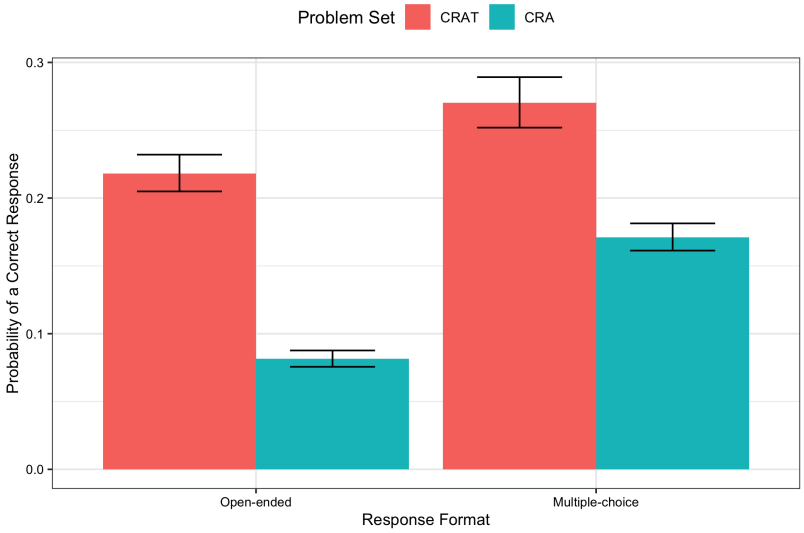


Figure S23: Effects of response format and item type on CRA scores

interaction. (In practical terms, this also suggests that the main effects analysis is probably not an accurate representation of the true relationships tested.) Table S21 shows the effects with the same reference groups (open-ended response format and CRAT problems). The odds ratios are 0.28 for the effect of response choice format and 1.33 for the effect of item content/type. Figure S23 shows a plot of the model including the interaction effect.

3 Study 3

The aim of Study 3 was to validate the new CRA item set from Study 1 in an additional (non-student) sample. In this case, the measures were cross-validated against measures of cognitive reflection (using items from various versions of the Cognitive Reflection Task; Frederick, 2005; Primi, Morsanyi, Chiesi, Donati, & Hamilton, 2016; Thomson & Oppenheimer, 2016) and imagination (using the Four-Factor Imagination Scales; Zabelina & Condon, 2020). This sample was also used to evaluate test-retest reliability. Data were collected from a crowd-sourcing recruitment platform (Prolific) in February/March 2021 (see dates below). These data are available on OSF (tinyurl.com/2a8vx6n8) in a file labeled 'Study3data.csv'; as these data are considerably more involved than those in Study 2, we have also provided a codebook. In the following subsections, we describe the steps for loading and cleaning the data, and carrying out the planned analyses.

3.1 Data wrangling

Begin by loading, cleaning, and scoring the data. Note that the data have already been de-identified, and that responses to the individual items have already been scored as correct (1) or incorrect (0).

```
rm(list=ls()[!(ls() %in% c('findcolnumber', 'keys.list', 'wdpath', 'score_ICAR_IRT', 'CRAitemNumbersStudy1'))])

# Load the de-identified data file. The raw data file has been cleaned to
# remove identifying data that is automatically collected by Qualtrics.

# Note that the data for Study 3 are very involved; see the codebook file on OSF
# for more info about the nature and naming of variables (`Study3dataCodebook').
study3data <- read.csv(paste(wdpath, "data/study3/study3data.csv", sep=""), header=TRUE)
study3data <- study3data[-c(1:2),]

# change class of variables in character format (not sure why... qualtrics)
cols.num <- colnames(study3data[,c(findcolnumber(study3data, "FFIS_F1"):
                     findcolnumber(study3data, "FFIS_D5"),
                     findcolnumber(study3data, "MultipleChoice_DO_q_14066_retest"):
                     findcolnumber(study3data, "MultipleChoice_DO_q_14051_retest"))])
study3data[cols.num] <- sapply(study3data[cols.num],as.numeric)

# clean out unexpected handling of NAs for some cases in qualtrics
study3data[cols.num] <- psych::scrub(study3data[cols.num], isvalue= c(-99))

names(study3data)[3] <- 'duration' # relabel
study3data$duration <- as.numeric(study3data[,"duration"]) # as numeric from character
study3data$age <- as.numeric(study3data[,"age"]) # as numeric from character
NbeforeDrops <- dim(study3data)[1]

study3data <- subset(study3data, select = -c(RecordedDate, UserLanguage)) # drop as uninformative

# Subset the data to remove respondents who self-report lower English fluency levels
table(study3data$english)

##
##                                          -99 Very well (fluent/native)                      Well
##                          5                   5                     361                        13

study3data <- study3data %>% dplyr::filter(english == "Well" | english == "Very well (fluent/native)")
dim(study3data)

## [1] 374 503

# Fix the date formatting
study3data[,"StartDate"] <- substr(study3data[,"StartDate"], 1, 7) # reformat the date columns
study3data[,"StartDate"] <- as.Date(study3data$StartDate, "%m/%d/%y")
study3data[,"EndDate"] <- substr(study3data[,"EndDate"], 1, 7) # reformat the date columns
study3data[,"EndDate"] <- as.Date(study3data$EndDate, "%m/%d/%y")
study3data[,"StartDate_retest"] <- as.Date(as.numeric(study3data$StartDate_retest), origin = '1900-01-01')
study3data[,"EndDate_retest"] <- as.Date(as.numeric(study3data$EndDate_retest), origin = '1900-01-01')
dim(study3data)

## [1] 374 503

# Subset the data to remove data collected outside the Prolific dates
study3data <- study3data[study3data$StartDate >= "2021-02-23",]
dim(study3data)

## [1] 357 503

Nactual <- dim(study3data)[1]

# When were the data collected in the first wave?
table(study3data$EndDate)

##
## 2021-02-23 2021-02-25
##         28        329

begin <- min(study3data$EndDate)
end <- max(study3data$EndDate)

# And when in the test-retest?
table(study3data$EndDate_retest)
```

```
##
## 2021-03-11 2021-03-12 2021-03-13
##         91          6          1

beginRetest <- min(study3data$EndDate_retest)
endRetest <- max(study3data$EndDate_retest)

study3data$retestInterval <- as.numeric(difftime(study3data$EndDate_retest, study3data$EndDate))
retestIntervalStats <- describe(study3data$retestInterval)
```

The first round of data collection for Study 3 ran from 2021-02-23 and ended on 2021-02-25; the second round (re-test data) were collected a mean of 14.7 days later (min = 14.3; max = 17.2). Of the 384 participants involved in the first wave, 27 were removed due to low self-reported levels of English language fluency. This left 357 participants at the first wave. From these, 98 participated in the retest assessment roughly two weeks later.

3.2 Descriptive statistics

The next code chunk describes the sample (N = 357) and key variables before moving on to the analyses.

```
descriptives[c("age", "FFISfrequencyScores", "FFISNegEmoValenceScores", "FFIScomplexityScores",
               "FFISdirectednessScores", "CRTscores", "CRAscoreIRTwave1_score",
               "CRAscoreIRTwave2_score"),c("n","mean","sd", "median", "min", "max")]

##                            n  mean    sd median   min   max
## age                      357 34.17 12.61  31.00 18.00 78.00
## FFISfrequencyScores      310  3.53  1.19   3.56  1.00  6.00
## FFISNegEmoValenceScores 310  2.85  1.10   2.86  1.00  5.71
## FFIScomplexityScores     310  2.91  1.09   2.80  1.00  6.00
## FFISdirectednessScores   310  3.72  1.00   3.60  1.20  6.00
## CRTscores                357  0.39  0.29   0.40  0.00  1.00
## CRAscoreIRTwave1_score   357 47.92  5.04  48.98 32.15 56.13
## CRAscoreIRTwave2_score    98 55.99  5.36  55.99 41.69 70.02

# Note that nearly all participants reported being fluent in English and almost all of these
# were native English speakers
table(study3datatrimmed$english_native)

##
##  No Yes
##  16 341

table(study3datatrimmed$english)

##
## Very well (fluent/native)                      Well
##                       347                        10

english <- data.frame(table(study3datatrimmed$english)[c(2,1)])
colnames(english) <- c("English fluency", "Participants")
english$Percentage <- round(english$Participants/sum(english$Participants)*100,1)

edu <- data.frame(table(study3datatrimmed$education)[c(7,6,3,8,1,2,4,5)])
colnames(edu) <- c("Educational Attainment Level", "Participants")
edu$Percentage <- round(edu$Participants/sum(edu$Participants)*100,1)

income <- data.frame(table(study3datatrimmed$income)[c(10,4,7,8,9,1,2,3,5,6,11,12)])
colnames(income) <- c("Household Income", "Participants")
income$Percentage <- round(income$Participants/sum(income$Participants)*100,1)

gender <- data.frame(table(study3datatrimmed$gender))
colnames(gender) <- c("Gender Identity", "Participants")
gender$Percentage <- round(gender$Participants/sum(gender$Participants)*100,1)
```

Descriptive statistics for the scale scores are shown in the code above. Note that the IRT scores for the CRA items (for both wave 1 and the retest values) range are T-scores based on the IRT parameters from Study 1. The scale scores for FFIS are mean scored, based on the items answered; a subset of the sample was not administered the FFIS items due to an complication in the coding of the survey administration

(some participants' response sessions timed out before they got to the FFIS section). Internal consistencies for the FFIS scales and the Cognitive Reflection Task, reported in Table S22, were all above threshold.

Scale	alpha	items
FFIS Frequency	0.93	9
FFIS Emotional Valence	0.91	7
FFIS Complexity	0.84	5
FFIS Directedness	0.81	5
Cognitive Reflection Task	0.85	10

Table S22: Internal consistency reliabilities for Study 3 scales

Descriptive statistics for the demographic characteristics of the sample are shown below in Tables S23 through S26. The sample contains a high majority of participants self-reporting English fluency, and 63.6% reported having an attained an associate's degree level of education or higher. About 77% of the sample reported being in a household with income of $100,000 or less. With respect to gender identity, 2.8% identified as non-binary, and the remainder of the sample was roughly split between those who endorsed "female" (49.6%) and "male" (47.3%); 1 participant preferred not to respond.

English fluency	Participants	Percentage
Well	10	2.8%
Very well (fluent/native)	347	97.2%

Table S23: English fluency in the Study 3 sample

Educational attainment level	Participants	Percentage
Less than 12 years	1	0.3%
High school graduate/GED	36	10.1%
Currently in college/university	27	7.6%
Some college/university, but did not graduate	66	18.5%
Associate degree	35	9.8%
College/university degree	131	36.7%
Currently in graduate or professional school	13	3.6%
Graduate or professional school degree	48	13.4%

Table S24: Educational attainment in the Study 3 sample

3.3 Validity and two-week test-retest analyses

Correlations among the key variables/scale scores are shown in the code below.

```
scoresSubsetStudy3 <- subset(study3datatrimmed, select = c(CRAscoreIRTwave1_score, CRAscoreIRTwave2_score,
                                                           CRTscores, FFISfrequencyScores, FFISNegEmoValenceScores,
                                                           FFIScomplexityScores, FFISdirectednessScores))
scoresCorStudy3 <- cor.ci(scoresSubsetStudy3, plot = FALSE)
scoresCorStudy3

## Call:corCi(x = x, keys = keys, n.iter = n.iter, p = p, overlap = overlap,
##     poly = poly, method = method, plot = plot, minlength = minlength,
##     n = n)
##
```

Household income (annual)	Participants	Percentage
Less than 20,000	48	13.4%
20,000 to 40,000	66	18.5%
40,000 to 60,000	70	19.6%
60,000 to 80,000	60	16.8%
80,000 to 100,000	32	9%
100,000 to 120,000	22	6.2%
120,000 to 150,000	24	6.7%
150,000 to 200,000	17	4.8%
200,000 to 250,000	5	1.4%
250,000 to 350,000	0	0%
350,000 to 500,000	2	0.6%
More than 500,000	2	0.6%
Prefer not to say	9	2.5%

Table S25: Household income in the Study 3 sample

Gender identity	Participants	Percentage
Female	177	49.6%
Male	169	47.3%
Non-binary	10	2.8%
Prefer not to say	1	0.3%

Table S26: Gender identity in the Study 3 sample

```
##  Coefficients and bootstrapped confidence intervals
##                          CRAIRT1 CRAIRT2 CRTsc FFISfS FFISN FFIScS FFISdS
## CRAscoreIRTwave1_score    1.00
## CRAscoreIRTwave2_score    0.73    1.00
## CRTscores                 0.44    0.31    1.00
## FFISfrequencyScores      -0.04    0.13    0.08  1.00
## FFISNegEmoValenceScores  0.00    0.12    0.00  0.38   1.00
## FFIScomplexityScores    -0.12   -0.11   -0.04 -0.24   0.30  1.00
## FFISdirectednessScores  -0.21   -0.28   -0.10  0.23  -0.09 -0.16   1.00
##
##  scale correlations and bootstrapped confidence intervals
##                 lower.emp lower.norm estimate upper.norm upper.emp    p
## CRAIRT1-CRAIRT2      0.65       0.65     0.73       0.81      0.79 0.00
## CRAIRT1-CRTsc        0.36       0.36     0.44       0.50      0.50 0.00
## CRAIRT1-FFISfS      -0.16      -0.16    -0.04       0.08      0.07 0.48
## CRAIRT1-FFISN       -0.09      -0.11     0.00       0.11      0.12 0.95
## CRAIRT1-FFIScS      -0.22      -0.24    -0.12       0.00      0.00 0.05
## CRAIRT1-FFISdS      -0.31      -0.32    -0.21      -0.09     -0.08 0.00
## CRAIRT2-CRTsc        0.14       0.14     0.31       0.48      0.47 0.00
## CRAIRT2-FFISfS      -0.05      -0.09     0.13       0.33      0.35 0.24
## CRAIRT2-FFISN       -0.02      -0.05     0.12       0.30      0.30 0.16
## CRAIRT2-FFIScS      -0.25      -0.28    -0.11       0.09      0.10 0.29
## CRAIRT2-FFISdS      -0.40      -0.43    -0.28      -0.11     -0.13 0.00
## CRTsc-FFISfS        -0.04      -0.04     0.08       0.19      0.19 0.20
## CRTsc-FFISN         -0.11      -0.12     0.00       0.11      0.11 0.92
## CRTsc-FFIScS        -0.13      -0.14    -0.04       0.07      0.07 0.55
## CRTsc-FFISdS        -0.21      -0.21    -0.10      -0.01      0.00 0.04
## FFISfS-FFISN         0.28       0.28     0.38       0.48      0.49 0.00
## FFISfS-FFIScS       -0.34      -0.35    -0.24      -0.12     -0.13 0.00
## FFISfS-FFISdS        0.12       0.10     0.23       0.34      0.34 0.00
## FFISN-FFIScS         0.20       0.20     0.30       0.40      0.40 0.00
## FFISN-FFISdS        -0.22      -0.22    -0.09       0.01     -0.01 0.07
## FFIScS-FFISdS       -0.30      -0.30    -0.16      -0.03     -0.03 0.02
```

Note that the correlations above include the two-week test-retest correlation ($r = 0.73$; 95% empirical CI: [0.65,0.79]). This association is also shown in Figure S24.

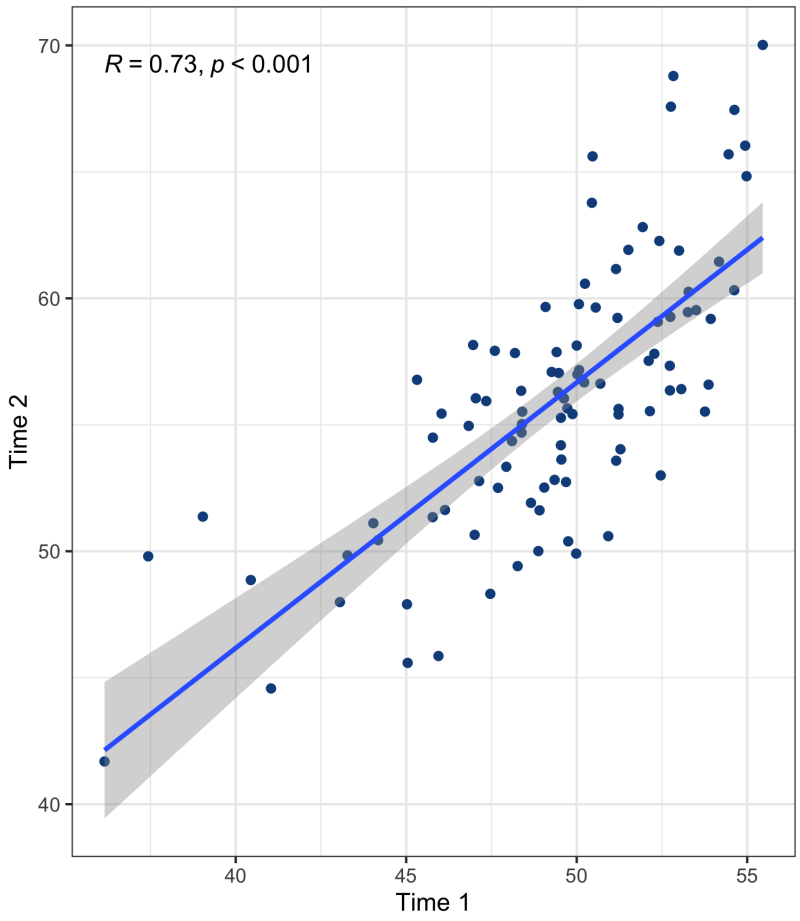


Figure S24: Test-retest reliability over two weeks

4 Study 4

4.1 Dimensionality and IRT analyses of the combined CRA item sets

The aim of Study 4 is to identify the IRT calibrations of a larger item set for use when administering the CRA with multiple choice response options in the future. The method matches the same procedures used in Study 1, however the item pool is expanded. Here, we generate calibrations for a set that combines the "new" CRA items and the multiple-choice version of the CRAT items from Study 1 (87 in total, after removing the problematic items), plus multiple-choice versions of an additional 80 items commonly used as open-ended items in CRA-related research (based on personal correspondence with Mark Beeman). Thus, **there are a total of 167 items in the combined CRA item pool reported on here**.

Of the 80 new items that we not previously used in Studies 1 or 2, 49 are part of the 144-item set described in Bowden and Jung-Beeman (2003) (these items are numbered q_14327 through q_14404, with many of the chronological item numbers excluded), and the remaining 31 items were algorithmically generated using the procedures described in Study 1. These 31 items have not previously been published, and are numbered q_14434 through q_14466. Note that items q_14442 and q_14437 were deprecated due to having an aberrantly low difficulty and discrimination values, respectively.

In the code version of this file (see the .Rnw), we begin by wrangling the data as follows: (1) loading this larger data file that overlaps with the one used in Study 1, (2) removing the items identified as problematic in Study 1 (2 items from the new 67-item set of the CRA and 8 from the older 30-item CRAT set), (3) removing the two items mentioned above, and (4) generating descriptive statistics.

The resulting data frame includes 167 CRA items and 1,502,922 participants. With respect to the overlapping nature of these data relative to Study 1, note that the Study 1 data were collected between 2017-02-07 and 2022-06-25. Here, we have added to those data by administering items more intensively from the larger item pool. Thus the data collection in this sample ran from 2017-02-07 to 2022-12-12. Over much of this interval, a small number of items were given from about half of the item pool (i.e., the data from Study 1), but a larger number of items were given from a larger item pool over the last 170 days. We use all available data to maximize our sample size — this is particularly important when identifying the IRT parameters.

The total number of administrations per item ranged from 4,542 to 28,872 (m = 14,151; Mdn = 19,125; sd = 9,325). The pairwise administration counts ranged from 153 to 1,275 (m = 304; Mdn = 209; sd = 212.8). In the next chunk, we identify the item-level correlation matrix (tetrachoric) and evaluate the unidimensionality.

```
tetchor_matrix167 <- suppressWarnings(tetrachoric(CRAitemSet167, smooth = TRUE))
```

```
cor_matrix167 <- tetchor_matrix167$rho

omega_results167 <- suppressWarnings(psych::omega(cor_matrix167, plot = FALSE))
omega.h167 <- round(omega_results167$omega_h,2)
omega.alpha167 <- round(omega_results167$alpha,2)
omega.t167 <- round(omega_results167$omega.tot,2)
```

The items have high internal consistency in this data set (α = 0.98; ω_t = 0.99). Similarly, the omega general value (ω_h = 0.85) suggests strong unidimensionality. To help visualize this, we plot the correlations among items after they have been sorted by factor loading (code not shown), with a one factor model specified (sorting by factor loading is only done to organize the presentation of items). We also extract the empirically-observed confidence intervals around the tetrachoric correlations here (also not shown; see the Rnw version).

The correlations are shown in Figure S25. The plot provides evidence of a clear general factor, though there are a small number of correlations between items with point estimates below 0 — as low as -0.168 — indicated with pale red tones in the lower right corner of the figure (average overall r = 0.27). The range of the 95% confidence intervals around these correlations (i.e., the range from the high to low end of the confidence interval) had a mean of 0.21 (minimum = 0.14 and maximum = 0.22). Of the 13,861 pairwise correlations, 1116 had confidence intervals overlapping with zero; only 9 of the remaining 12,745 had point estimates *and* confidence intervals below 0.

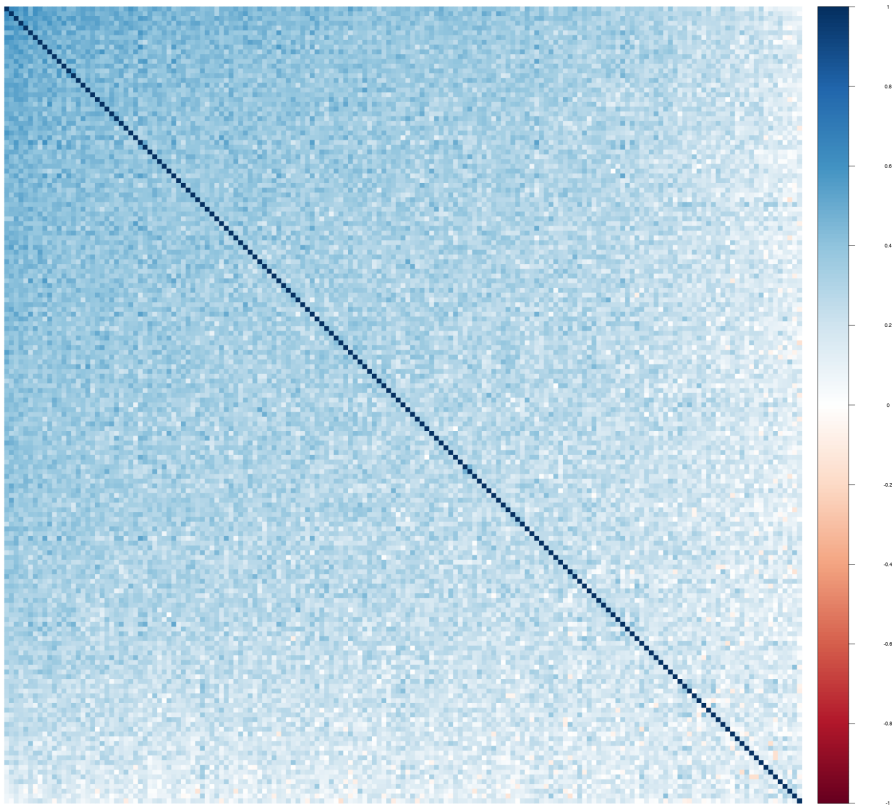


Figure S25: Correlations among 167 multiple choice CRA items sorted by general factor loadings.

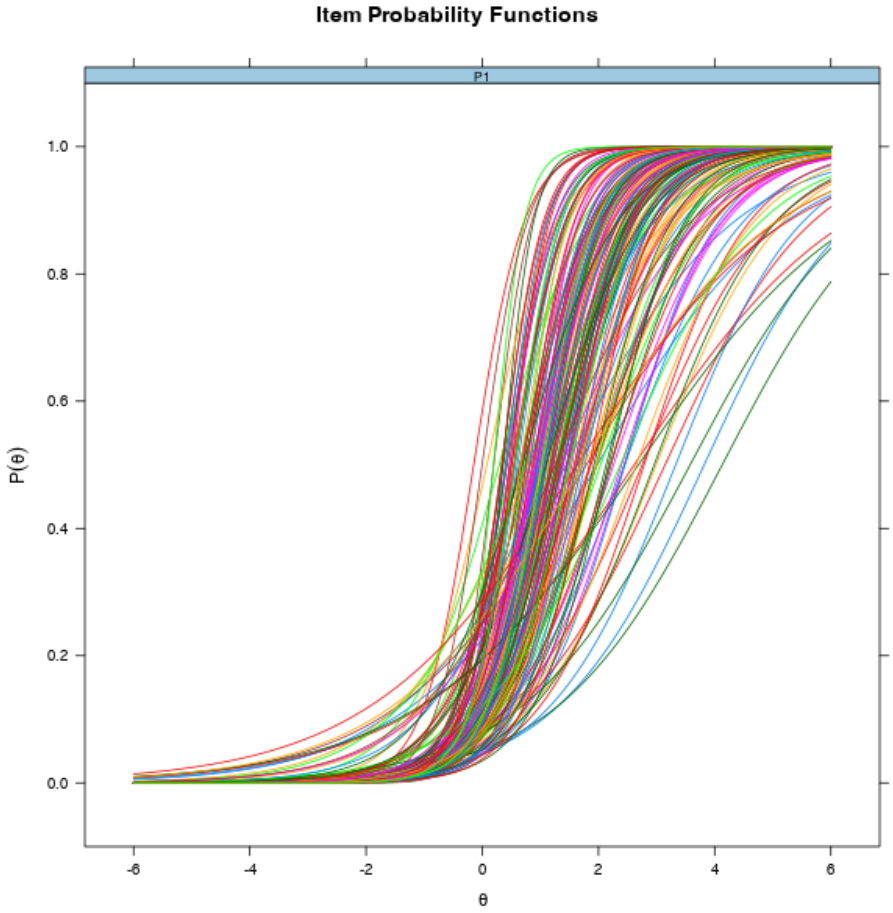


Figure S26: Item probability functions for all of the CRA items together.

4.2 Identify the IRT parameters for the full CRA item bank

The next step is to extract and table the IRT parameters for the full item bank, then plot the item characteristic curves (for all items separately and together) and test information functions. The code for these steps is left out of the PDF version of this file (see the Rnw for full details). Table S27 shows the items and parameters. This includes coefficients for the item difficulties (the location of the 50/50 probability of a correct response) and discrimination parameters (the slopes), as well as the mean proportion of correct responses and the standard error of the means.

Figure S26 shows the probability functions for all the items together in one plot, and Figure S27 shows the probability functions for all 167 items separately. Figure S26 indicates that most items have difficulties greater than zero ($m = 1.33$; $Mdn = 1.21$). Difficulty values reflect the location of the intercept where the probabilities of correct and incorrect responses are equal. Only 2 of the 167 items have difficulty coefficients below zero, and this indicates that most items are somewhat difficult for most respondents. Given the content of these items, moderately high difficulties are desirable, but items with excessively high difficulty would not be particularly useful as there will be little variance across respondents (i.e., because nearly all respondents fail to identify the correct answer).

Figures S26 and S27 also show that most of the items are adequately discriminant ($m = 1.67$; $Mdn = 1.65$), though there are a few exceptions. This implies that the items are able to discriminate well between individuals with relatively similar levels of associative ability around the difficulty intercept, as a small

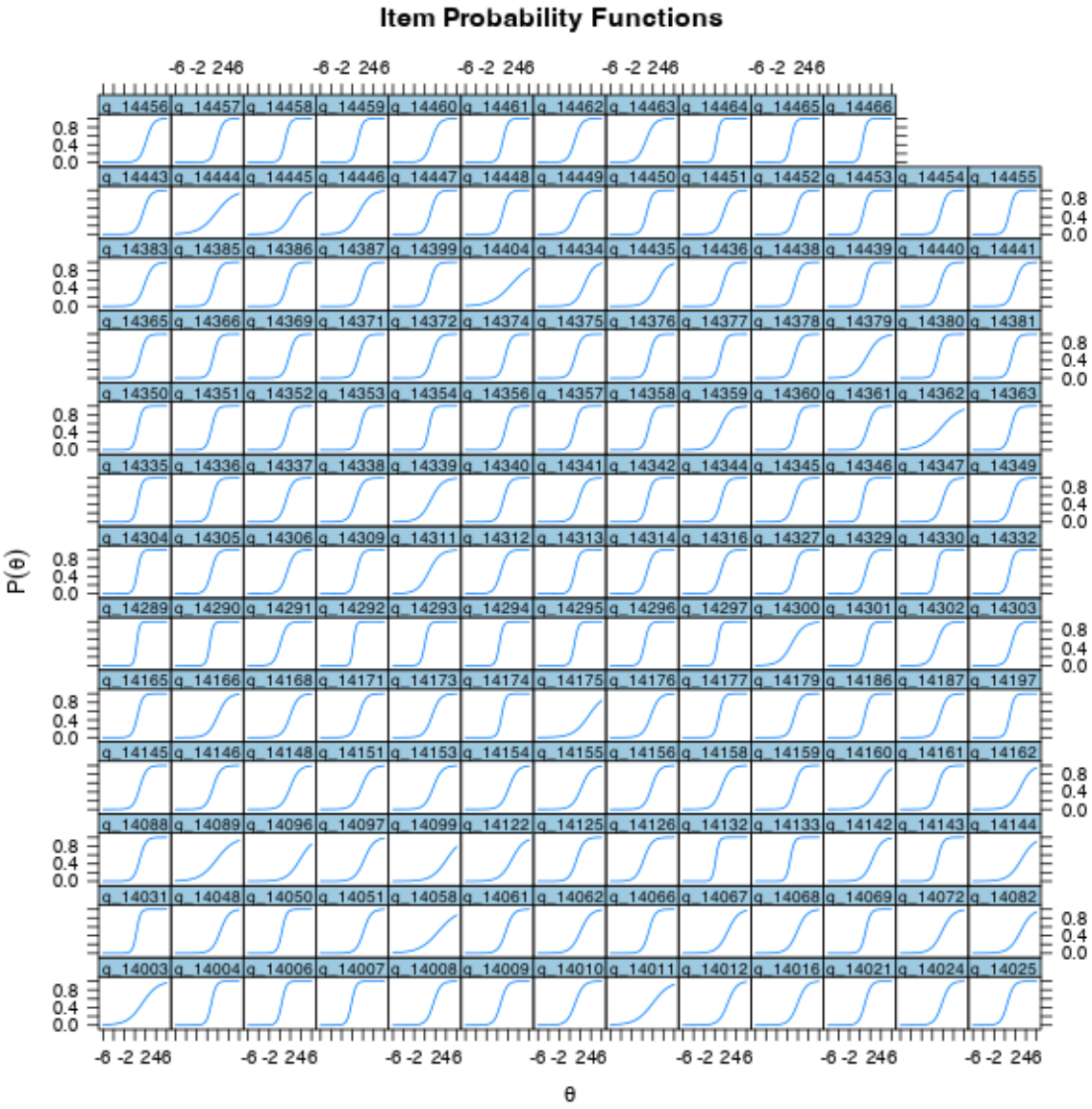


Figure S27: Item probability functions for all of the CRA items.

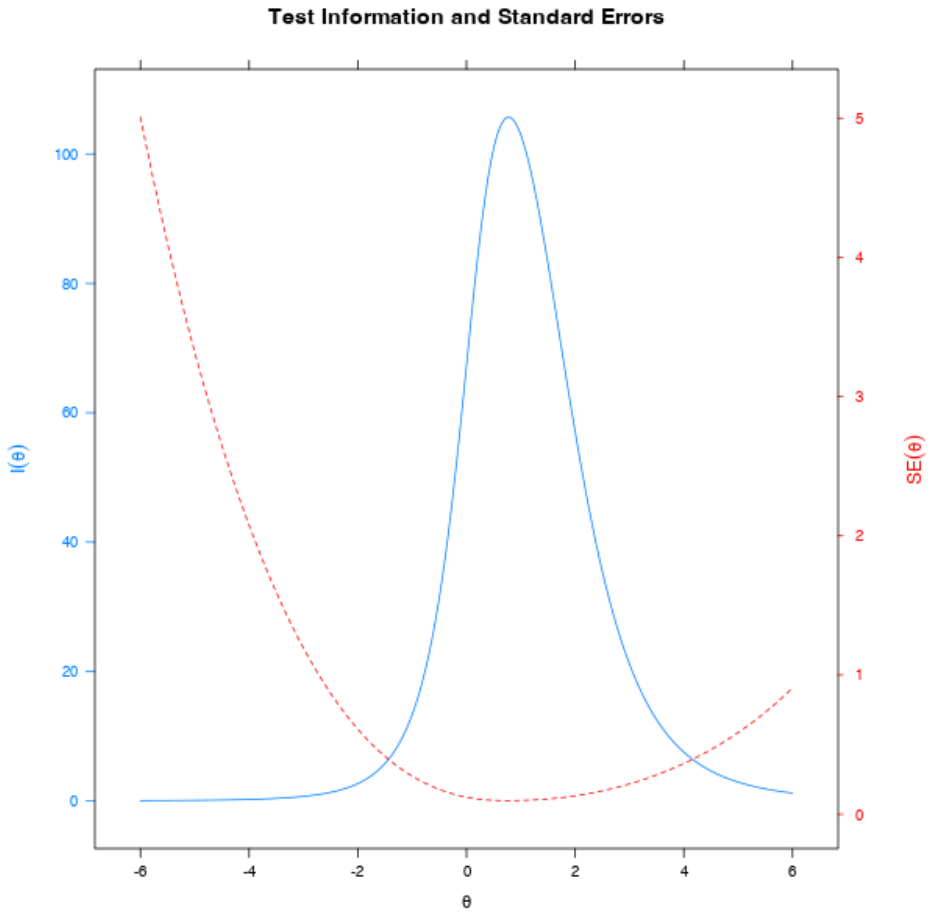


Figure S28: Test information (blue) and standard error (red) functions for the complete CRA item pool.

increase in associative ability in this range substantially increases the probability of correct responding. This is depicted visually by the slope of the item probability functions. Of the 167 items, only 3 have discrimination parameters below .6 and above .5, a value that is commonly deemed a lower bound of acceptability (An & Yung, 2014; Debeer & Janssen, 2013). Note that it may also be reasonable to drop a few other items with relatively low discriminations, but there is a case to be made for keeping items that cover a wide range of discriminations, especially if they are all above an acceptable threshold (Cheng & Liu, 2015). This argument mainly rests on the increased flexibility provided by a diverse item set for computer adaptive testing applications.

Figure S28 shows the joint information for the final set of CRA items across the range of difficulty along with the standard error. As the standard error is akin to the inverse of reliability (more specifically, reliability is 1 minus the square of the standard error over the standard deviation), Figure S28 indicates that the test is particularly reliable from the range of difficulty of about -0.5 sd to +3 sd. The marginal reliability across all respondents in these data is high at 0.96 (Thissen & Wainer, 2001).

The parameters for the (2PL) IRT model for the full item bank — shown in Table S27 — are posted for use by others OSF with the rest of the materials for this project; see tinyurl.com/2a8vx6n8.

Table S27: Item parameters for complete CRA item pool

Item	Discrimination	Difficulty	Mean	SE	N
q_14003	0.70	1.44	0.57	0.003	28412
q_14004	2.20	0.77	0.51	0.003	28563
q_14006	2.32	0.36	0.48	0.003	28196
q_14007	2.32	-0.15	0.44	0.004	19465
q_14008	1.48	1.64	0.44	0.003	28279
q_14009	1.64	0.64	0.44	0.004	19988
q_14010	1.29	1.02	0.41	0.004	19392
q_14011	0.62	1.98	0.40	0.003	19802
q_14012	1.19	2.44	0.38	0.003	20289
q_14016	1.23	1.97	0.37	0.003	28277
q_14021	1.76	0.99	0.37	0.003	28080
q_14024	1.38	1.86	0.37	0.003	19370
q_14025	1.64	0.50	0.37	0.003	28269
q_14031	2.69	0.33	0.37	0.007	4908
q_14048	1.24	2.45	0.36	0.003	19938
q_14050	2.28	0.40	0.36	0.003	28429
q_14051	1.43	2.13	0.35	0.003	28433
q_14058	0.55	2.62	0.35	0.003	28163
q_14061	1.42	1.61	0.34	0.003	28430
q_14062	1.05	2.04	0.33	0.003	19822
q_14066	1.67	1.22	0.33	0.003	20008
q_14067	0.98	2.45	0.33	0.003	28382
q_14068	1.10	1.87	0.33	0.003	28033
q_14069	2.11	1.11	0.32	0.007	4961
q_14072	0.97	1.86	0.31	0.007	4693
q_14082	0.92	2.99	0.31	0.007	5045
q_14088	1.72	0.93	0.30	0.003	28027
q_14089	0.61	1.79	0.30	0.007	4696
q_14096	0.79	3.82	0.30	0.003	28019
q_14097	1.08	2.26	0.29	0.006	4978
q_14099	0.70	4.13	0.29	0.007	4862
q_14122	0.88	2.75	0.29	0.003	19078
q_14125	1.45	1.29	0.29	0.007	4542
q_14126	1.30	0.29	0.28	0.007	4598
q_14132	3.07	0.20	0.28	0.003	19782
q_14133	2.54	0.57	0.28	0.006	4838
q_14142	1.12	2.40	0.28	0.003	28861
q_14143	1.97	1.01	0.27	0.006	5288
q_14144	0.79	3.15	0.27	0.003	28159
q_14145	1.58	0.94	0.27	0.007	4645
q_14146	1.44	1.10	0.27	0.003	19399
q_14148	1.18	1.96	0.27	0.003	20007
q_14151	1.15	1.69	0.27	0.003	28491
q_14153	1.21	1.60	0.27	0.006	4881
q_14154	1.23	2.22	0.27	0.003	28872
q_14155	1.13	2.20	0.27	0.006	4784
q_14156	1.40	1.73	0.26	0.003	19335
q_14158	1.24	1.25	0.26	0.003	19713
q_14159	1.63	1.29	0.25	0.003	19832
q_14160	0.92	3.33	0.25	0.006	4758
q_14161	1.59	0.98	0.25	0.006	4555
q_14162	0.96	2.96	0.25	0.006	4846
q_14165	2.06	0.50	0.25	0.003	19536
q_14166	1.07	1.97	0.25	0.006	5486
q_14168	1.31	1.97	0.24	0.006	4971
q_14171	1.51	1.21	0.24	0.006	5342
q_14173	1.31	1.38	0.24	0.003	19460
q_14174	2.61	0.82	0.24	0.003	19902
q_14175	0.69	3.58	0.24	0.006	4628
q_14176	1.27	1.63	0.24	0.003	19865
q_14177	2.25	0.80	0.24	0.003	20070
q_14179	1.68	1.36	0.23	0.003	28311
q_14186	2.05	0.94	0.23	0.003	19600
q_14187	1.50	1.46	0.23	0.006	4807
q_14197	2.34	0.93	0.23	0.002	28557
q_14289	3.79	0.35	0.23	0.006	4569
q_14290	3.06	0.47	0.23	0.003	20118
q_14291	1.66	0.08	0.23	0.006	5257
q_14292	3.80	0.19	0.23	0.006	5472
q_14293	2.58	-0.00	0.22	0.002	28419
q_14294	2.84	0.40	0.22	0.006	4636
q_14295	2.62	0.88	0.21	0.006	4869

(continued on next page)

Table S27: Item parameters for complete CRA item pool

Item	Discrimination	Difficulty	Mean	SE	N
q_14296	2.28	0.43	0.21	0.003	19125
q_14297	3.07	0.50	0.21	0.006	5404
q_14300	0.86	0.84	0.21	0.006	4807
q_14301	2.16	1.15	0.21	0.003	19550
q_14302	1.35	0.82	0.20	0.003	19476
q_14303	1.69	1.30	0.20	0.006	4799
q_14304	2.64	0.47	0.20	0.003	19258
q_14305	1.78	0.73	0.20	0.003	20119
q_14306	1.47	1.38	0.20	0.006	4598
q_14309	2.02	0.80	0.20	0.006	4682
q_14311	0.92	0.76	0.20	0.003	20024
q_14312	1.59	0.64	0.20	0.006	4671
q_14313	2.15	0.93	0.20	0.006	4824
q_14314	1.70	0.89	0.19	0.005	5222
q_14316	1.61	1.10	0.19	0.005	5227
q_14327	1.91	1.29	0.19	0.003	19263
q_14329	1.78	1.11	0.19	0.006	4767
q_14330	2.65	0.71	0.19	0.005	5436
q_14332	2.22	0.90	0.19	0.006	4848
q_14335	2.42	0.61	0.18	0.002	28638
q_14336	1.90	0.92	0.18	0.002	28441
q_14337	1.54	1.18	0.18	0.005	5308
q_14338	1.49	0.66	0.18	0.002	28279
q_14339	0.92	1.14	0.18	0.005	5206
q_14340	1.76	1.16	0.18	0.006	4617
q_14341	1.47	1.67	0.18	0.005	4893
q_14342	1.97	0.96	0.18	0.006	4782
q_14344	1.75	1.17	0.18	0.006	4775
q_14345	1.54	1.27	0.18	0.003	19639
q_14346	2.28	0.54	0.18	0.003	20130
q_14347	1.46	1.95	0.18	0.005	5432
q_14349	1.56	1.28	0.17	0.003	19739
q_14350	2.28	0.56	0.17	0.005	5350
q_14351	2.06	0.81	0.17	0.003	20041
q_14352	1.80	1.40	0.17	0.005	5404
q_14353	1.92	1.45	0.17	0.005	4682
q_14354	2.43	0.75	0.17	0.003	20150
q_14356	1.93	1.59	0.17	0.005	4798
q_14357	1.96	0.91	0.16	0.003	18916
q_14358	1.93	1.20	0.16	0.005	4565
q_14359	1.04	1.02	0.16	0.002	28827
q_14360	1.95	1.54	0.16	0.005	4619
q_14361	1.49	1.47	0.16	0.005	4670
q_14362	0.56	1.64	0.15	0.003	19646
q_14363	1.83	1.16	0.15	0.005	5128
q_14365	1.84	1.04	0.15	0.003	19313
q_14366	1.98	1.17	0.14	0.005	4949
q_14369	1.73	1.77	0.14	0.005	4631
q_14371	1.68	1.22	0.14	0.002	19980
q_14372	1.57	1.33	0.14	0.002	19585
q_14374	1.88	1.50	0.14	0.005	5430
q_14375	1.67	1.03	0.14	0.005	5192
q_14376	1.88	1.34	0.14	0.005	5109
q_14377	2.01	1.10	0.14	0.002	20870
q_14378	1.65	1.38	0.14	0.005	5276
q_14379	0.85	1.37	0.14	0.005	5095
q_14380	2.02	1.07	0.14	0.002	19136
q_14381	1.89	1.35	0.13	0.005	5320
q_14383	1.33	1.85	0.13	0.005	5243
q_14385	1.68	1.26	0.13	0.002	28751
q_14386	1.65	1.31	0.13	0.005	4822
q_14387	1.66	1.72	0.13	0.005	4797
q_14399	1.77	0.93	0.13	0.005	5299
q_14404	0.53	2.69	0.13	0.002	19817
q_14434	1.11	2.78	0.12	0.005	4606
q_14435	0.99	2.65	0.12	0.004	5348
q_14436	1.58	1.49	0.12	0.002	18987
q_14438	1.73	1.46	0.12	0.005	4608
q_14439	2.02	0.98	0.12	0.002	20218
q_14440	1.56	1.46	0.11	0.002	19992
q_14441	1.51	1.92	0.11	0.002	19757
q_14443	1.50	1.58	0.11	0.005	4586
q_14444	0.61	1.73	0.11	0.002	19910

(continued on next page)

Table S27: Item parameters for complete CRA item pool

Item	Discrimination	Difficulty	Mean	SE	N
q_14445	0.83	2.38	0.11	0.004	4768
q_14446	0.86	1.26	0.11	0.004	5369
q_14447	1.67	1.11	0.10	0.002	28227
q_14448	1.82	1.07	0.10	0.004	5408
q_14449	1.21	1.10	0.10	0.002	19718
q_14450	1.91	0.86	0.09	0.002	20195
q_14451	1.41	1.68	0.09	0.002	19449
q_14452	1.67	1.51	0.09	0.004	4718
q_14453	2.05	0.69	0.09	0.004	5357
q_14454	1.75	1.62	0.09	0.004	5217
q_14455	2.07	0.88	0.09	0.002	20028
q_14456	1.53	2.17	0.09	0.004	4827
q_14457	1.75	1.85	0.09	0.004	5337
q_14458	2.13	1.01	0.08	0.002	20061
q_14459	2.10	1.05	0.08	0.002	19761
q_14460	1.40	1.28	0.08	0.002	27898
q_14461	2.14	1.06	0.08	0.002	19509
q_14462	1.54	0.93	0.07	0.004	4677
q_14463	1.23	0.69	0.06	0.002	19735
q_14464	2.82	0.34	0.06	0.003	5419
q_14465	2.38	0.66	0.06	0.002	19382
q_14466	2.58	0.56	0.06	0.002	19407

Notes: The values for mean, se (standard error of the mean), and n are based on individual administrations; the discrimination and difficulty values are based on a 2PL item response model.

4.3 A mini-tutorial based on the IRT parameters for the full CRA item bank

In the final sub-section, we use the IRT parameters identified in the prior section to re-visit the data sets in Studies 2 and 3. This is done for two reasons: (1) to show how the parameters can be used to produce generalizable scale scores that allow for the comparison of scores across samples, and (2) to compare the samples used in the current work. Recall that the data in Study 1 are highly overlapping with the data used to derive the IRT parameters; Studies 2 and 3 are 'out-of-sample' in this case.

The procedures are relatively straight-forward. We need (a) the scoring function, (b) the scoring parameters, and (c) the data to be scored. We have been using the IRT scoring function throughout this project — it was actually included in the very first code chunk, visible in the .Rnw version of the file. For transparency, we print the code of that function here for reference; it is essentially a wrapper around the `fscores()` function in the mirt package (Chalmers, 2012).

```
rm(list=ls()[!(ls() %in% c('findcolnumber', 'keys.list', 'wdpath', 'score_ICAR_IRT'))])

score_ICAR_IRT

## function(dataset, IRT_model, SE = TRUE, t_score = TRUE)
## {
##   factor = deparse(substitute(IRT_model))
##   factor = gsub("IRT_model_", "", factor)
##   items <- names(IRT_model@Data$mins)
##   data <- dataset
##   #check items in dataset
##   check_names = items %in% names(dataset)
##   # if missing all items, throw error
##   if(sum(check_names) == 0) stop("No items for this factor found in the dataset.")
##   # if missing only some items, add those items as empty columns
##   if(sum(check_names) < length(check_names)){
##     missing = items[!check_names]
##     data[,missing] = NA
##   }
##   data <- data[, items]
##   # find rows that are completely empty. these cause problems with mirt
##   empty_rows = which(is.nan(rowMeans(data, na.rm = TRUE)))
##   if(length(empty_rows) > 0) data = data[-empty_rows, ]
##   # Calculate IRT factor scores:
##   fscores <- fscores(
```

```
##     object = IRT_model,
##     response.pattern = data
##     )
##   fscores <- as.data.frame(fscores)
##   scores = fscores[,1]
##   if(t_score) scores = scores*10 + 50
##   if(t_score) fscores[,2] = fscores[,2]*10
##   if(length(empty_rows) > 0){
##     # if empty rows
##     dataset[-empty_rows, factor] = scores
##     dataset[-empty_rows, paste0("SE_", factor)] = fscores[,2]
##   }else
##   {
##     # If there are NOT empty rows:
##     dataset[, factor] = scores
##     dataset[, paste0("SE_", factor)] = fscores[,2]
##   }
##   return(dataset)
## }
## <bytecode: 0x7f7d8a1e0e30>
```

The scoring parameters were identified in the previous subsection. As previously stated, the model can be downloaded directly from the OSF page (tinyurl.com/2a8vx6n8); look for the file named 'IRTmodelCRAitemsFullBank.rdata'. In the next code chunk, we show it being loaded (after being downloaded).

```
load(paste(wdpath, "data/study4/IRTmodelCRAitemsFullBank.rdata", sep = ""))
```

Then we load the data for the studies and trim the data frames to focus on the CRA items only. These data have previously be transformed to 1s and 0s to reflect correct or incorrect responses, respectively.

```
# Study 2
load(paste(wdpath, "data/study2/CRAdataStudy2.rdata", sep = ""))
CRAdataStudy2 <- subset(study2dataCRAonly, select = c(CRAitemNumbersStudy2))
dim(CRAdataStudy2)

## [1] 684  63

# Recall that 63 items were left at the end of Study 2

# Study 3
load(paste(wdpath, "data/study3/CRAdataStudy3.rdata", sep = ""))
CRAdataStudy3 <- subset(study3dataCRAonly, select = c(CRAitemNumbersStudy3))
dim(CRAdataStudy3)

## [1] 357  66

# Recall that 66 items were left at the end of Study 3
```

Now, use the IRT scoring function and IRT parameters to score the data for each study and then plot to compare the scores across samples.

```
# Study 2
CRAstudy2scores = score_ICAR_IRT(dataset = CRAdataStudy2, IRT_model = IRT_model_ICAR_CRA, t_score = FALSE)

# Study 3
CRAstudy3scores = score_ICAR_IRT(dataset = CRAdataStudy3, IRT_model = IRT_model_ICAR_CRA, t_score = FALSE)
```

```
file_path = paste(wdpath, "figure/comparesamples.png", sep = "")
png(height=600, width=1200, file=file_path, type = "cairo")
plot1 <- ggplot(CRAstudy2scores, aes(x=ICAR_CRA)) +
  geom_histogram(bins = 100) + labs(x="CRA IRT-based scores") +
  lims(x = c(-2,2), y = c(0,25))
plot2 <- ggplot(CRAstudy3scores, aes(x=ICAR_CRA)) +
  geom_histogram(bins = 100) + labs(x="CRA IRT-based scores") +
  lims(x = c(-2,2), y = c(0,25))
grid.arrange(plot1, plot2, ncol=2)
dev.off()

## pdf
##   2
```

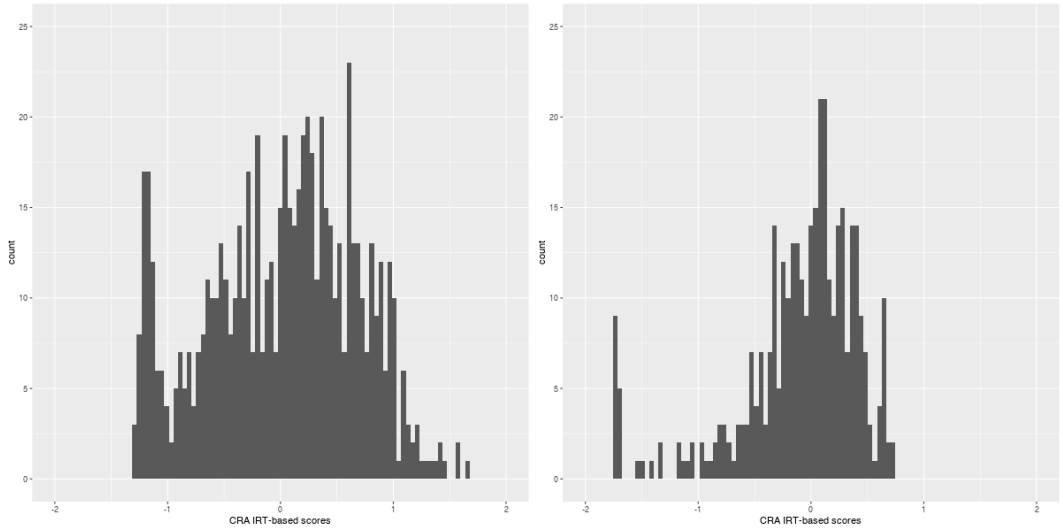


Figure S29: Distribution of CRA scores in Study 2 (left) and Study 3 (right) based on IRT scores for the full item bank

Figure S29 shows the distributions of IRT scores for the CRA scales in Studies 2 and 3. The undergraduate sample from Study 2 (the distribution on the left) has slight evidence of a floor effect but the scores are mainly distributed across the range of difficulty. The online sample from Study 3 has a restricted range on the high end, but a similarly broad range of scores overall. Though it is not possible to know what factors contribute to low scores in either sample, it seems very likely that low scores are influenced, at least in part, by low engagement with the CRA items. In sum, it appears that the IRT parameters of the CRA item bank provide a substantially wide range of scores in samples like those that are typically used in psychology research, without much restriction of range in either direction. In combination with the other aspects of reliability and validity evaluated in the prior studies, this provides strong evidence of the utility of the CRA item bank for the assessment of associative ability.

FIN

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